CONCEPTUAL MEDICINE

FINAL AID

By

Dr. Paras Jamwal

DAY-1

FINAL AID

FMGE RAPID REVISION

Chanting-Style High-Yield Review

- Anatomy
- Physiology
- Biochemistry
- Pathology
- Microbiology
- Pharmacology



19 SUBJECTS | 03 DAYS

DAY-2

FINAL AID

FMGE RAPID REVISION

Chanting-Style High-Yield Review

- PSM
- Forensic Medicine
- ENT
- Ophthalmology
- Orthopaedics
- Anaesthesia



19 SUBJECTS | 03 DAYS

DAY-3

FINAL AID

FMGE RAPID REVISION

Chanting-Style High-Yield Review

- Internal Medicine
- Surgery
- Obstetrics & Gynaecology
- Paediatrics
- Radiology
- Dermatology



19 SUBJECTS | 03 DAYS

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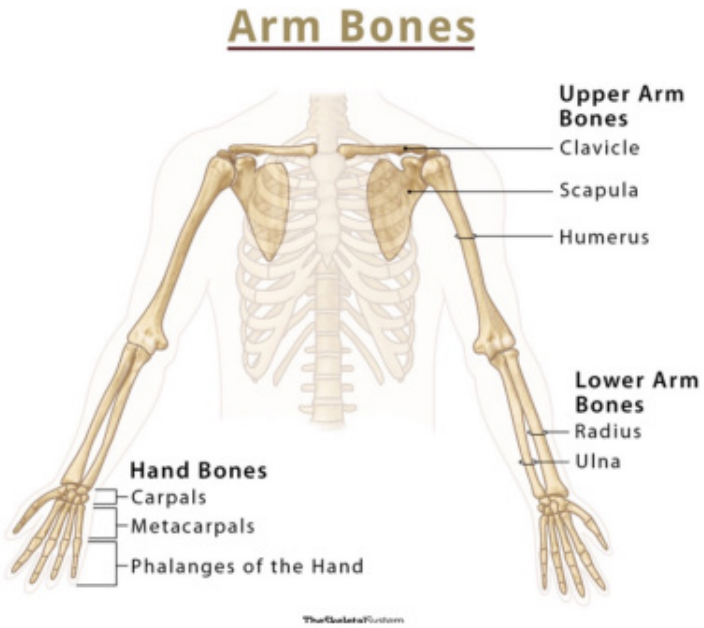
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ANATOMY

Upper Limb

- Mc Deformities
  - → Ape Thumb → Median Nerve
  - → Partial Claw Hand → Ulnar Nerve
  - → Complete Claw Hand → Ulnar + Medial Nerve
  - → Tenni's Elbow → Lateral Epicondylitis
  - → Golfer's Elbow → Median Epicondylitis
  - → Wrist Drop → Radial Nerve
  - → Foot Drop →Common Perional Nerve
  - → Labourer's Nerve → Median Nerve
  - → Musician Nerve → Ulnar Nerve



- Humerus
  - → Surgical Neck - Axillary Nerve
  - → Shaft - Radial Nerve
  - → Post. To Medial Epicondyle - Ulnar Nerve
- Hand
  - → Largest Carpel bone → Capitate
  - → Smallest Carpel bone → Pisiform
  - → 1st to Ossify → Capitate
  - → Last to Ossify → Pisiform
  - → Mc # Bone → Scaphoid
  - → Mc Bone to go under AVN → Scaphoid
  - → Lunat Dislocation → Median Nerve Compression
- Muscles Of Rotator Cuff
  - S → Supra Spinatous
  - I → Infra Spinatous
  - T → Teres Minor
  - S → Subscapularis
- Imp. Root Value's
  - → Axillary Nerve - C5, C6
  - → Radial Nerve - C5, 6, 7, 8, T1
  - → Medial Nerve - C5, 6, 7, 8, T1
  - → Ulnar Nerve - C7, C8, T1
  - → Long Thoracic Nerve → C5, C6, C7
- Winging of Scapula
  - → Injury to long thoracic nerve
  - → Serratus ant. is paralysed

Imp. Muscles

I. Serratus Anterior = Boxer's Muscle

Nerve - Long Thoracic Nerve

Action - Pushing & Punching

II. Latissimus Dorsi = Climber's Muscle

Nerve - Thoracodorsal Nerve

III Pectoralis Major

Nerve - Medial & Lateral Pectoral nerve

Action - Flexion, Adduction & Medial rotation

IV Deltoid

Nerve - Axillary Nerve

Action - Abduction at shoulder joint

↳ Abduction of shoulder joint

0- 15° - Supra Spinatous

15-90° - Deltoid

> 90° - Serratus Anterior + Trapezius

↳ Axillary Nerve Injury

- Deltoid & Teres Minor paralysed

- Loss Of Abduction

- Regimental Badge Sensory Loss - Due to upper lateral cutaneous nerve which isbranch of axillary nerve

V Trapezius

Nerve - Spinal Accessory Nerve

Action - Upper Fiber → Elevation of scapula

Middle Fiber → Retroction of scopula

Lower Fiber → Depression of scapula

VI Muscle of ant. compartment of arm

B → Bicep's Brachi

B → Brachialis

C → Coracobrachialis

Nerve - Musculo Cutaneous nerve

Action - Flexion of arm

VI Muscle of post compartment of arm

→ Triceps

Nerve - Radial Nerve

Action - Extension of arm

- Erbes Palsy -

→ C5, C6 roots are involved

→ Policeman tip deformity seen

→ Limbs are in → Adduction

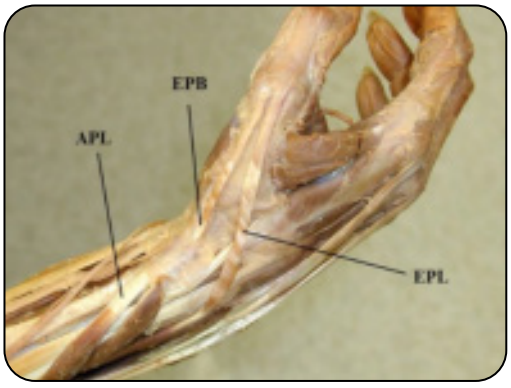
Medial Rotation

Pronation

- Erbes Palsy -
  - → C5, C6 roots are involved
  - → Policeman tip deformity seen
  - → Limbs are in → Adduction
    Medial Rotation
    Pronation

- Klumpke's Palsy
  - → C8, T1 roots are involved
  - → Full claw hand is seen
  - → Horner syndrome seen
    - ↳ P - Ptosis
    - ↳ M - Miosis
    - ↳ A - Anhydrous
    - ↳ L - Loss of ciliospinal reflex
    - ↳ E - Enophthalmos

- Dermatome's Of Upper Limb
  - → Thumb - C6
  - → Middle 3 Fingers - C7
  - → Little Finger - C8



- Anatomical Snuff Box
  - ↳ Ant. Lateral Boundary
    - → APL - Abductor Pollicis Longus
    - → EPL - Extensor Pollicis Brevis

↳ Post. Medial Boundary

EPL - Extensor Pollicis Longus

↳ Floor

→ S - Scaphoid

→ T - Trapezium

↳ Roof

→ Cephalic vein

→ Superficial branch of radial nerve

↳ Content

→ Radial Artery

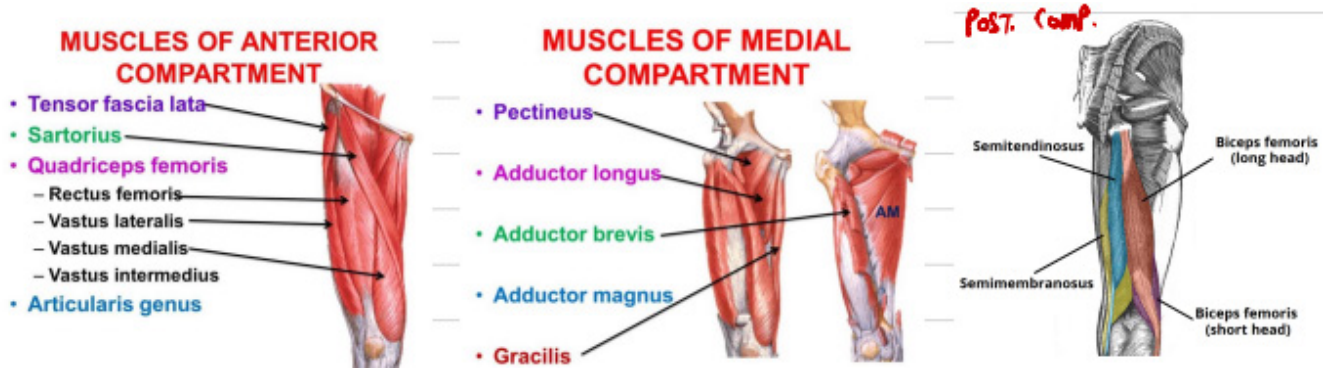
- Supination & Pronation occur at → Radio - Ulnar Joint
- Dorsal Interosi → Abduction of finger's
- Palmar Interosi → Adduction of finger's
- Lower border of Scapula → C7

Lower Limb

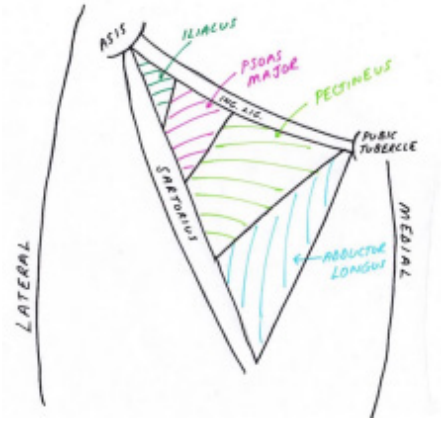
- Thigh

Ant. Compartment	Post. Compartment	Medial Compartment
↓	↓	↓
Femoral Nerve L2, L3, L4	Sciatic Nerve L4,S3	Obturator Nerve L2,L3, L4
↓		↓
Post. Division Of Ventral Rami		Ant. Division Of Ventral Rami

- Muscle Of Ant. Compartment Of Thigh

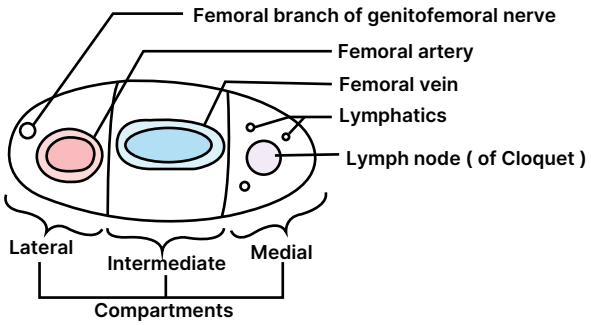


- Femoral Δ



- External Iliac Artery
  - ↓ Pass through deep inguinal liq
- Femoral Artery
  - ↓ Pass through adductor hiatus
- Popliteal Artery

- Femoral Sheath



- Gluteal Region

A → Gluteus Maximus

Nerve - Inf. gluteal nerve

Action - Extensor of hip joint

- Helps in standing & sitting

B → Gluteus Minimus & Medius

Nerve - Superior gluteal nerve

Action - Chief abductor of hip joint

Injury to Superior Gluteal Nerve

↳ Lurching GAIT

- Dorsi flexion & Planter flexion occur at ankle joint
- Eversion & inversion occur at subtalar joint

- Sciatic Nerve

Tibial Nerve
→ Nerve of post. comp. of leg

Common Peroneal Nerve
Wind around neck of fibula

Deep Peroneal Nerve
→ Nerve of ant. comp of leg

Superficial Peroneal Nerve
→ Nerve of lateral comp. of leg

- 1= Web space is supplied by → Deep Peroneal Nerve

Thorax

- **RIB's**
  - → True RIB - 1-7
  - → False RIB - 8-12
  - → Floating RIB - 11, 12
- Sternum Angle
  - ↳ Angle b/w manubrium & body of sternum
  - ↳ +nt at lower border of T_4
- Neuro Vascular Bundle
  - V - Vein
  - A - Artery
  - N - Nerve

  } +nt at lower border of rib
- Heart
  - → All veins of heart drains into coronary sinus except ant. cardiac vein, it drains directly into right atrium
  - → Inf. Surface - Both Ventricles
  - → Base - Both Atria
  - → Apex - Left Ventricles
- Trachea
  - → 10 - 12cm length
  - → From lower border of cricoid cartilage C_6 - T_6

- Esophagus
  - → 25cm length
  - → Lower border of C_6 - C_{11}
  - → Narrowest Point - C_6 [Cricopharyngeal Junction]

- Diaphragm

Opening	Level	Structure passing
Vena Canal opening	T8	IVC Rt. Phrenic nerve
Oesophageal opening	T10	Oesophagus Vagal Trunk Left Gastric Artery
Aortic opening	T12	Aorta Thoracic Duct Azygous Vein

Head & Neck

Cranial Nerves

1 → Olfactory Nerve
2 → Optic Nerve
3 → Oculomotor Nerve
4 → Trochlear Nerve

5 → Trigeminal

> Maxillary Nerve

> Ophthalmic Nerve

> Mandibular Nerve

6 →Abducent Nerve

7 →Facial Nerve

8 → Auditory Nerve

9 →Glossopharyngeal Nerve

10 → Vagus

11 →Accessory Nerve

12 →Hypoglossal Nerve

- Imp. Foramens Of Skull

→ Cribriform Plate - CN1

→ Optic Canal - CN2 + Ophthalmic Artery

→ Foramen Rotandum - Maxillary Nerve

→ Foramen Ovale - Mandibular Nerve
Accessory Nerve
Lesser Petrosal Nerve
Emissary Vein

- Nerve supply of face
  - Sensory supply of face

    ↳ Full face is supplied by trigeminal nerve

    Except → Skin over angle of mandible supplied by great auricular nerve
  - Motor supply of face

    Full face is supplied by fascial nerve

    Except → Muscle of mastication supplied by mandibular nerve
  - Muscle of mastication

    Internal Carotid Artery

    → Masseter

    → Temporalis

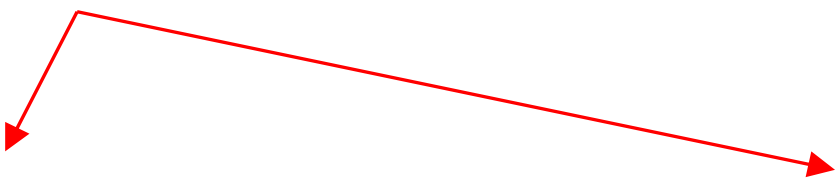
    → Medial Pterygoid

    → Lateral Pterygoid

    Supplied by mandibular nerve

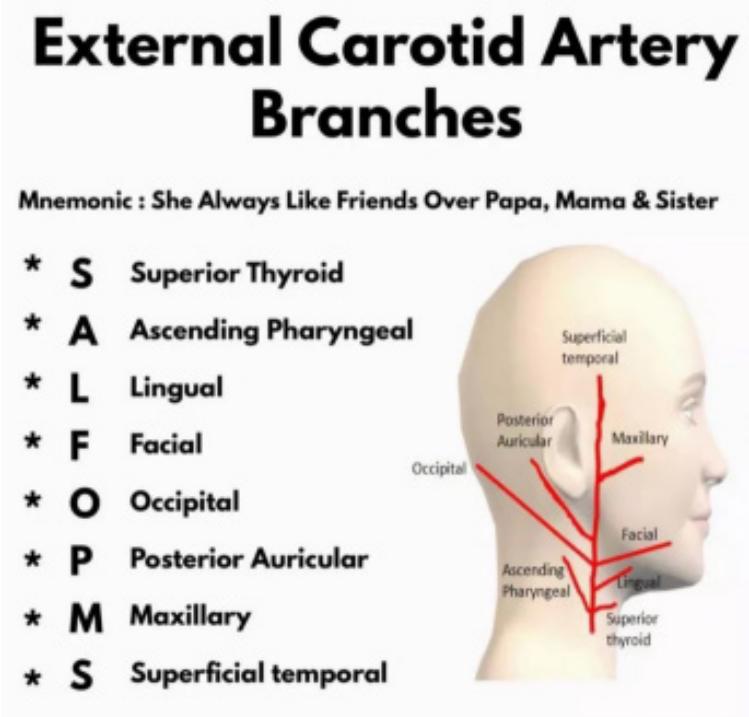
    - Depression of mandible/ opening of mouth

      ↳ τ the help of lateral pterygoid
  - Common Carotid Artery



External Carotid Artery

Internal Carotid Artery



Fascial Artery

↳ Superior Labial Artery

↳ Inferior Labial Artery

↳ Tonsillar Artery

Maxillary Artery

↳ Middle Meningeal Artery

↳ Sphenopalatine Artery

↳ Greater Palatine Artery

- Salivary Gland

  I Parotid Gland

  ↳ Drained by-Stenson's Duct/ Parotid Duct

  ↓

  Open near upper 2nd molar tooth

- Frey's Syndrome [Mostly during parotid gland Sx.]

  ↳ Due to damage to auriculo temporal nerve

  Which is branch of mandibular nerve

S/s in Frey Syndrome

→ Redness, Flushing, Sweating

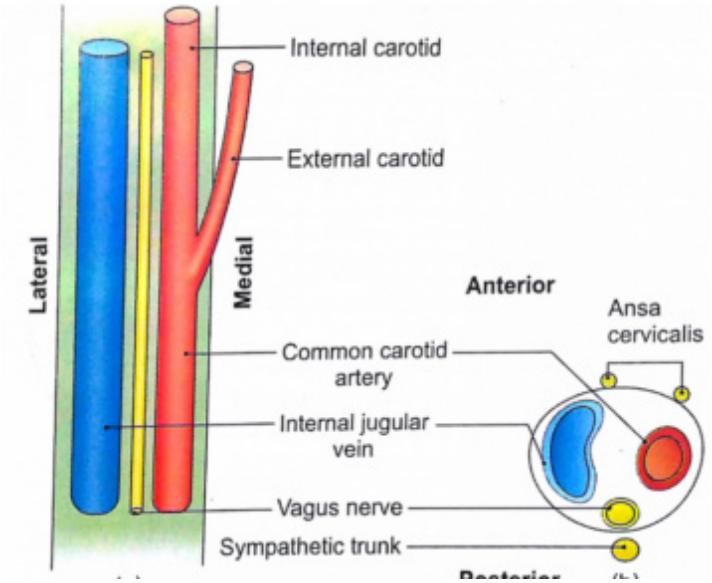
→↑ Temp. on lateral aspect of cheek while eating

II Submandibular Gland

↳ Drained by - Wharton's Duct

↳ Opens at - Floor of oral cavity at Sublingual Papilla

- Carotid Sheath



→ Muscle supplied

- Sternohyoid
- Omohyoid
- Sternothyroid

- **Tongue**

↳ Motor Supply

→ All muscles of tongue are supplied by hypoglossal nerve except palatoglossus which is supplied by Vagoaccessory complex [CN 10 & 11]

Sensory Supply

→ Ant. 2/3 part

↳ General Sensation - By Lingual Nerve

↳ Taste Sensation - By Chorda Tympani

→ Post. 1/3 part

↳ General sensation
↳ Taste sensation
} Glosopharyngeal nerve

→ Post. Most part

↳ General sensation
↳ Taste sensation
} Vagus nerve

- Safety muscle of tongue → Genioglossus
- Hypoglossal N. injury → Tongue protude to same side of lesion

- **Littel's Area**

→ Antero inferior quadrant of nasal septum

→ Supplied by kisselbach's plexus

↳ Ant. ethmoidal artery

↳ Greater palatine artery

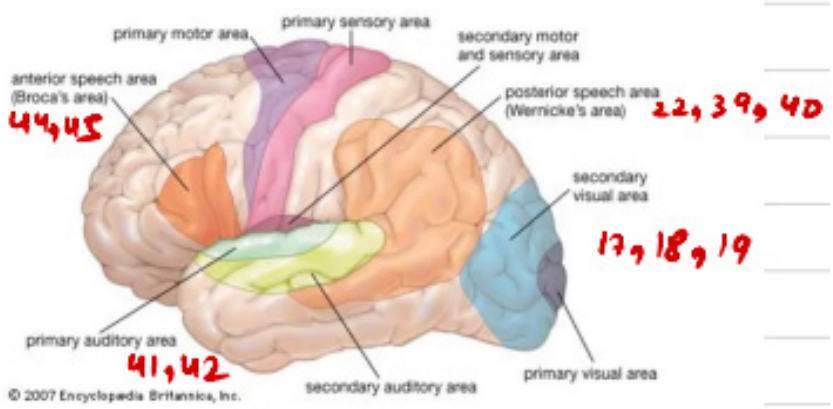
↳ Sphenopalatine artery

↳ Superior labial artery

- Cranial Nerve
  - ↳ Pure Sensory - CN 1, 2, 8
  - ↳ Pure Motor - CN 3, 4, 6, 11, 12
  - ↳ Mixed - CN 5, 7, 9, 12
    - Thickest/ largest cranial nerve - Trigeminal N.
    - Thinnest cranial nerve - Trochlear N.
    - CN τ longest intracranial course - Trochlear N.
    - CN τ longest extra cranial course - Vagus N.
    - CN τ longest intraosseus course - Facial N.

Neuro Anatomy

- Imp. brain area



Functional areas of the human brain.

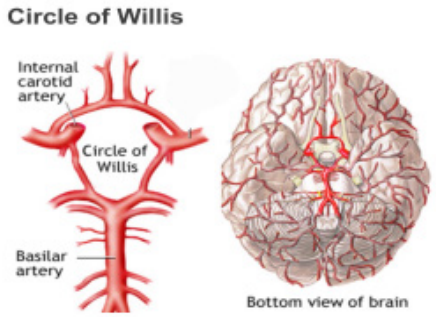
- Cavity in brain = Ventricle

  Lateral Ventricle → Cerebral Hemisphere

  3rd Ventricle → b/w Diencephalon

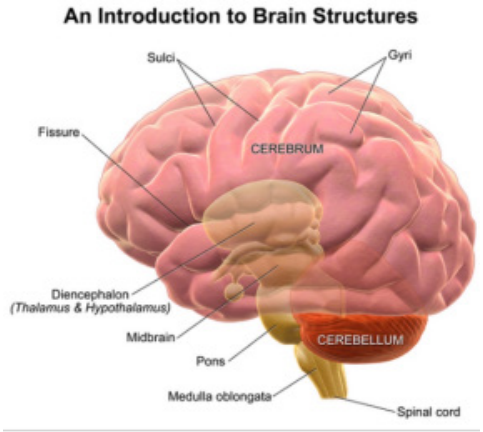
  4th Ventricle → b/w Pons & Cerebellum

- ACIA [Ant. inf. cerebellar artery] is a branch of basilar artery
- PICA [Post. inf. cerebellar artery] is a branch of vertebral artery



Formed by

- Ant. Cerebral Artery
- Post. Cerebral Artery
- Ant. Communicating Artery
- Post. Communicating Artery



- Parts of Basal Ganglia
  - P → Putamen
  - A → Amygdala
  - S → Substantia Nigra
  - S → Subthalamus
  - G → Globus Pallidus
  - C → Caudate
  - C → Claustrum

- Deep Cerebellar Nuclei

  D - Dentate Nucleus

  E - Emboli Form Nucleus

  F - Fastigial Nucleus

  G - Globase Nucleus

Abdomen & Pelvis

Abdominal aorta

↓

Common Iliac artery

Ext. Iliac artery Int. Iliac artery

→ Ventral Branches

↳ Coelic Trunk - Fore Gut ⟶ Left gastric artery, Common hepatic artery, Splenic artery

↳ Superior Mesenteric Artery - Midgut

↳ Inferior Mesenteric Artery- Hind Gut

→ Lateral Branches

↳ Gonadal artery

↳ Renal artery

↳ Mid. Suprarenal artery

↳ Inferior phrenic artery

→ Posterior Branches

↳ 4 pair of lumbar artery

↳ 1 median sacral artery

→ Terminal Branches

↳ Rt & Lt Common Iliac Artery

- Portal Vein

  Formed By → Union of splenic vein & superior mesenteric vein at LI

  - Policeman Of Abdomen - Greater Omentum
  - Bed Of Stomach - Never formed by spleen
  - Large Intestine

    ↳ Longest part - Transverse Colon

    ↳ Mc position of appendix - Retrocaecal

    ↳ Least common position of appendix - post-Ileal

- Duodenum

  ↳ 25cm

  ↳ Largest part - 3rd part

  ↳ Brunner's gland +Nt

- Calot Triangle - Content = Cystic Artery

  ↳ Inferior surface of Liver

  ↳ Common hepatic duct

  ↳ Cystic duct & gallbladder

- Common bile duct & pancreatic duct open
  -2nd parts of Duodenum
- True support of Pelvis - Round Ligament
- Descent of Testis
  → 4- 6th month - Deep Inguinal Ring
  → 7th month - Inguinal Canal
  → 8th month - Superficial Ing. Ring
  → 9th month - Scrotum

PHYSIOLOGY

CNS

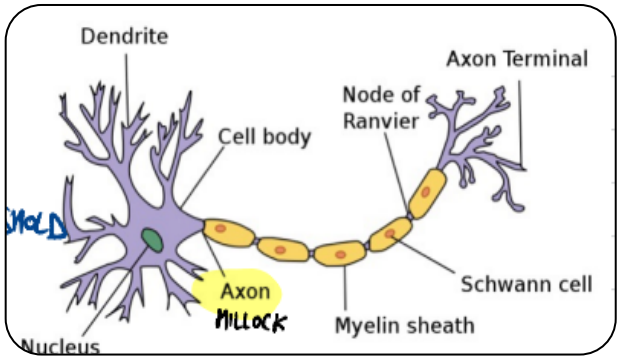
- **Glial Cell = 4types**

  1. Astrocyte - Make BBB
  2. Microvilli - Phagocytosis of bacteria
  3. Ependymer Cells - They line the ventricles
  4. Oligodendrocytes - Make myelin sheath in CNS but in PNS - Schwan cells makeMyelin

- **CSF**
  - 150 ml
  - Daily production = 500ml/ day
  - Secretion - Choroid Plexus
  - Absorption - Arachnoid Plexus

- **Neuron**

  → Axon Hilock =
  - AP generated due to min. threshold
  - Main ion = Ca^{2+}, Na^{2+}
  - RMP generated at rest
  - Main ION = K^{+}, Cl^{-}



RMP
- Neuron = -70mv
- Skeletal & Cardiac Muscle = -90mv

- Smooth Muscle = -50mv
- Rods & Cones = -30 To -40mv
- RBC = -12mv

- K & Cl^- makes membrane more -ve Hyperpolarization

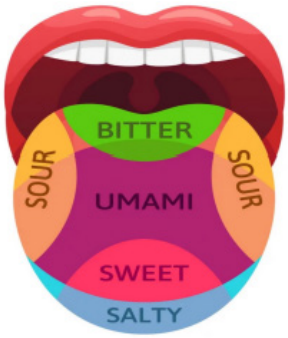
  Example - Gaba & Glycine (N.T)
- Na^+ & Ca^{2+} makes membrane less -ve Depolarization

  Example - Glutamate (N.T)
- Na+ & K+ pump play main role

  - Nokia - 3 Na^+ Out, 2K^+ In } both use ATP
  - Na+↑ Outside - Create AP
  - K+↑ Inside - Create RMP
- **Cortical Centre** = Helps in memory formation
  - Memory stored in synapses of neuron → Neocortex
  - ↑Memory due to ↑ neurotransmitter & ↑ receptor at synapse = Synaptic Plasticity
  - Short term memory is converted to long term memory - CA-1 neuron of Hippocampus

- Cortical Area
  - Auditory - 41, 42
  - Speech - 44, 45 k/as → Broca's Area
  - Language - 22 k/as → Wernicke's Area
- Taste

1. Salty = Na+/K+
2. Sour = H+/ Acid



3. Sweet = G-Protein Receptor
4. Bitter = G-Protein Receptor
5. Umami = Mono-Sodium Glutamate

- Chilly/ Spice = Not taste but pain sensation
  - ↳ Receptor - Vanilloid receptor
  - ↳ Sensation - Hot, burning, pain full sensation

- SSS - Somato Sensory System
  - Sensation from body - Skin, joints, muscle etc.
  - Pain, Temp. touch
- Somato sensory System receptor
  - 1. Meissner's Corpuscles
    - Tip of dermal papillae
    - Rapidly adapting, low threshold touch receptor

- Fine touch & 2-point discrimination
- Low frequency vibration

2. Pacinian Corpuscles
   - Onion shape
   - High frequency vibration in joints
   - Pressure in deep dermis & dermis
3. Merkel Disk's
   - Slow adapting touch receptor
   - Helps blind people to read

4. Ruffini Ending's → Warm
5. Kranse End Bulbs → Cold

- Slow adapting
- Sense- Pressure & Temperature

6. Free Nerve Ending
   - Pain, temp, sexual sensation & itching

   ↳ Pain -nt in brain & viscera → (Int. hollow organ)

   ↳ Pain +nt in outer covering

2 types of Pain

Fast Pain	Slow Pain
A delta myelinated somatic N. fibreeg-Skin Sharp & well localized	C- unmyelinated automatic N. fibreeg - Viscera (heart, intestine) Dull & diffused

- Ascending sensory tracts → Spinal Cord

  1. Anterior Spino Thalmic Tract = Carry itching & crude touch/ pressure

  2. Lateral Spino Thalmic Tract = Carry pain & temp

  3. Dorsal Column Tract = Rest all

- Nerve Injury

  - LMN Lesion
    ↓
  - Flaccid Paralysis
  - ↓Tone

  - UMN Lesion
    ↓
  - Spastic Paralysis
  - ↑ Tone

  - Muscle length sensed & maintained by muscle spindle = Stretch reflex/deep tendonreflex
  - Muscle tension sensed by golgi tendon organ
  - Cerebellum = Coordination & correction of movement
  - Basal Ganglia = Planning & programming of movement
  - Limbic System = Emotions genesis

- Frontal Lobe = Cognition & control of emotions

- **Hypothalamus Function :**

1. Control ANS → Sympathetic - Lat. Hypothalamus
   → Parasympathetic - Ant. Hypothalamus
2. Thirst centre → Trigger by Iblood & tsalt
   → Nerve = Lateral Preoptic Nerve
3. Circadian rhythm → By Suprachiasmatic Nerve
   (24 hrs Cycle) (int. Clock)
4. Hunger centre

Hunger	Satiety
N - Neuropeptide G - Ghrelin O - Orexins	Leptin

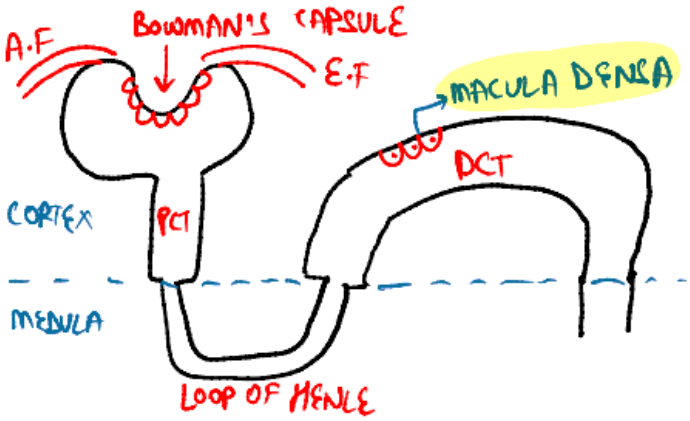
5. Temp. regulation



Anterior
- Cools body

Posterior
- Warms body

- **Kidney =**
  - Functional Unit → Nephron [1-1.3 million/ Kidney]



= Regulate GFR by tuboglomerular

  - Glomerular Capillaries (GC) =
    - Contain Endothelial Cells & Podocytes

↓ ↓

Openings = Endothelial Cell — Filtration Slitz

Size = 70 -90nm — 10-20nm

Filtration membrane for plasma

= 4-8nm

  - Size < 4 nm can easily filter
  - 4-8nm depends on charge [-ve charge CANT filter]

Example -> Albumin = Size 7nm but -vely charged

  - >8nm CANT filter

- **Urine formation has 3 steps**

1. Filtration = All substance get filtered & comes in PCT

2. Reabsorption [urine to blood] = Good solutes like Na+, Cl-, glucose, water, etc goback to blood by peritubular capillaries

3. Secretion [blood to urine]= Bad solutes like uric acid, drug's, toxin's,creatinine etc will come out from body with urine

- **PCT =**
  - Max. Reabsorption = Max reabsorption of all substance
    Except = Mg^{2+} inascending loop
  - Max Secretion = Max secretion of all substance
    Except = K^{+} & Urea
    ↓ ↓
    In CD & DCT In thin loop
- Medulla =
  High Osmolarity (Hypertonic) = 1200 mosm/L
  - Counter Current System

Multiplier → Ascending Loop → Transport Solutes

Exchanger → Vasa Reta Capillaries → Reabsorption Water

- **DCT**
  1. Na+ & Cl- Reabsorption
     - ↳ By Na^+ Cl^- symport
     - ↳ Blocked by Thiazide
  2. K+ Secretion By P-Cells [Principle Cells]
     - ↳ K^+ secretion & Na^+ reabsorption by ENAC channel
     - ↳ Blocked by amiloride
  - CD has 2 types of cells
    - ↳ P-cell → K^+ Secretion
    - ↳ I-cell → Acid Secretion & Make Urine Acidic
    - ↳ CD has V_2 receptor of ADH [Anti Diuretic Hormone]
      - ADH ↑ water reabsorption & make urine concentrated
      - If ADM is ↓
        - ↳ ↑ urine output = Polyuria
        - ↳ ↑ drinking = Polydipsia

        } Diabetes Insipidus
- Glucose =

  100% reabsorbed in PCT by SGLT-2 & GLUT-2
- TMG = Transport max. of glucose
  - It is max. glucose amount, that can be reabsorbed per min.
  - TMG = 375mg/ min

- Fl = Filter Load

  Amount getting filtered per minute

  In normal Person = TMG >> FL

  If FL>> TMG = Then some glucose is left unabsorbed & it comes in urine

  ↓

  Glycosuria

- Clearance = Amount of plasma cleared per min by kidney by filtration and secretion.

  CL = GFR : Only Filtration - Inulin

  CL > GFR : Filtration & Secretion -PAH [Para Amino Hippuric] & Creatinine

  CL < GFR : Filth Reabsorption - Glucose, AA

- GFR =

  Most accurate method → Inulin

  Mc used → Creatinine

- RPF = Renal Plasma Flow

  Estimated By PAH - Para Amino Hippuric Acid

- Juxta Glomerular Apparatus

  3 types of cells

  1. JG-cells → +nt in afferent arteriole

  2. Lacis cells → Helps in phagocytosis

  3. Macula Densa → Regulate GFR

- JG-cells → +nt in afferent arteriole
  - They sense renal blood flow = 1250ml/min
  - If flow is poor = Renin released
    ↓
    Raas activated

Respiratory System

→ Exchange Zone - Generation 17-23

→ Conduction Zone - Generation 0-16

→ Receptor H1 β2 M3
Dilation
Constriction
Constriction

→ Boyle's Law - $P \frac{1}{\propto} V$ Eg - Balloon

- Inspiration
  → Active process & need ATP for muscle cont.
  → Muscle used - Diaphragm 75% & ext. inter coastal 25%
- Expiration
  → Passive process & no A TP needed for any muscle
  → It is due to elastic recoil of the lung
- Force Full Expiration
  → Active process
  → Muscle used - Abdominal muscle & int. intercoastal

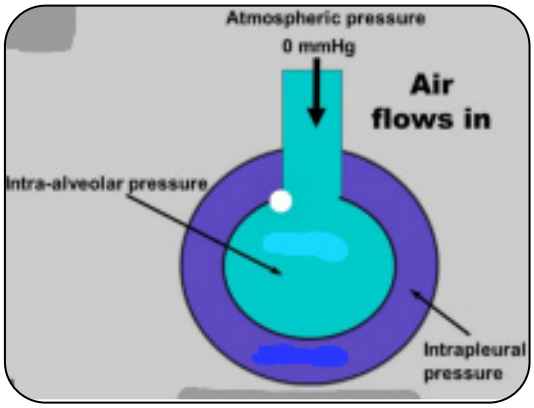
Q1 No Muscle Used For →

Q2 Into Intercostal Muscle →

Q3 Ext. Intercostal Muscle →

Q4 Abdominal Muscle →

Q5 Diaphragm →



- Intra Alveolar Pressure
  - → At Rest → 0 mmHg
  - → During Inspiration → -1 mmHg
  - → End Of Inspiration → 0 mmHg
  - → During Expiration → +1 mmHg

Steps Of Inspiration
1. Muscle Contract
2. Expand Thorax
3. -ve Pressure
4. Air Enter

- Intra Pleural Pressure
  - → At rest → -2 mmHg
  - → Normal quiet inspiration → -6 mmHg
  - → Normal quiet expiration → -2 mmHg
  - → Forceful inspiration → -20 mmHg
  - → Forceful expiration → +20 mmHg

- Compliance - $\frac{\Delta V}{\Delta P}$ Change in volume / Change in pressure
  [Stretching]

- RLD = Restrictive Lung Disease
  → Small lung bcz. no compliance
  Example → Lung Fibrosis

- Old = Obstructive Lung Disease

  → Big lung bcz. good compliance & poor recoil

  Example → Emphysema

- Surfactant

  → Produced by Type II Pneumocytes & Clara cell

  → Function - ↓ Surface Tension

  → 70% DPPC - Di Palmitoyl Phosphodil Choline

  ↳ Lecithin

  ↳ Sphingomylin

  → L/S Ratio → 2 normal

  → Production - Start at 18 wks of fetal life

  → Secretion & action inside alveoli - 28 wks

  → Max secretion & function - 34-35 wks

- Lung Volumes

  1. Tidal Volume - TV - 500ml
  2. Inspiratory Reserve Volume - IRV - 3-3.5L
  3. Expiratory Reserve Volume - ERU - 1.3L
  4. Residual Volume - RU - 1.2L

- Lung Capacities

  1. Inspiratory Capacity - IRV + TV

     - 3-3.5 + 500ml = 3.5 -4L

2. Expiratory Capacity - ERV +TV

- 1.3 + 500ml = 1.8L

3. Functional Residual Capacity - ERV + RV

- 1.3 + 1.2 = 2.5L

4. Vital Capacity - IRV + ERV + TV

Male - 4.8L

Female - 3.2L

Note - Spirometer cannot measure RV, so FRC also not given by spirometer

Bcz FRC = ERV + RV

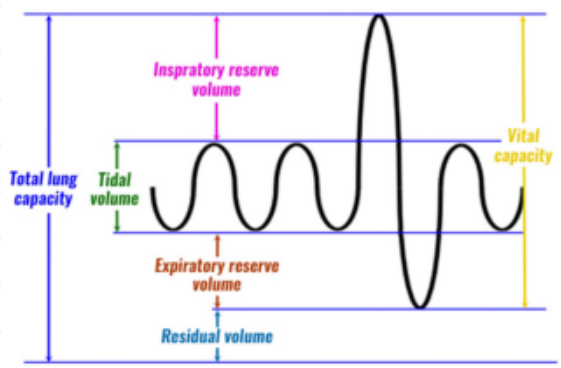
- FRC measured by
  1. Helium Dilution method
  2. Single breath nitrogen washout method
     or

     Fowler Method
  3. Plethysmography in pt. with OLD
- Best Ix. to diff. old from RLD - FEV_1 %

	VC	FEV1%	IRV	RV	
• RLD	↓↓	N	↓	&	FEV_1 % - 75-80% of TVC
• OLD	N/↑	↓	N/↑	↑	FEV_2 % 95% of TVC
					FEV_3 % - 98% of TVC

Total Lung Capacity

= IRV + TV + ERV + RV

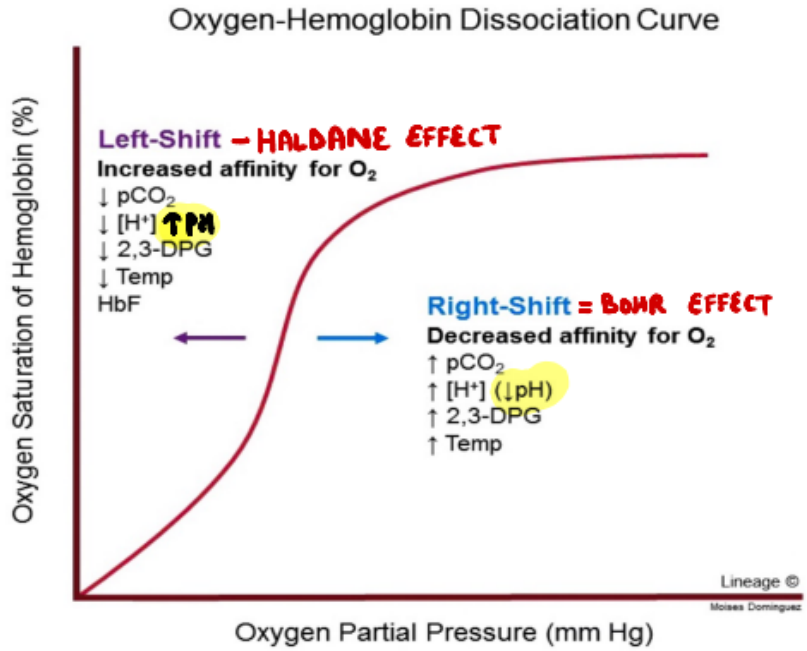


Air in anatomical dead space - 150ml

Air get exchanged - 350ml

- V/Q Ratio = Air coming for exchange / Blood coming for exchange = 1
  - → Ideal V/Q ratio - 1
  - → Normal V/Q ratio - 0.8
  - → Apex V/Q ratio - 1.3 = Max. = Poor Perfusion
  - → Base V/Q Ratio - 0.6 = Min. = High Perfusion
- Daltons Law For PP
  - → % of gas in a mixture X Total pressure
- Gas exchange depends on 3 factors
  - → Depends on membrane
  - → PP
  - → Gas
- Henry Law
  - → PaO2 depend on dissolved O_2 not Hb bound O_2

- Note -
  - → 1Hb can bind - 4 oxygen molecule
  - → 1gm Hb carry - 1.33ml oxygen
  - → Dissolved O_2 is only 1%
- Oxygen Hb dissociation curve



- Pace maker of respiration
  - → Pre Botzinger complex
    - Pons → Maintain respiration in proper way
    - Medula → Make Respiration
      - ↳ Medula Lesion - No respiration/ Apnea
      - ↳ Pons Lesion - Irregular respiration
- CO_2 act on both peripheral & central receptor
  - ↳ Main stimulus for regulation of respiration

CVS

- Pace Maker → SA Node → Max Frequency
- Sa Node can make 100AP in 60sec
- Av Node can make 60AP in 60sec
- Slowest conduction AV node - 0.05m/ sec
- Fastest conduction purkinjee fibres - 4m/sec

- Cardiac Cycle = 0.8sec

	Systole	Diastole
Atria	0.1 sec	0.7 sec
Ventricles	0.3 sec	0.5 sec

- Heart Sound

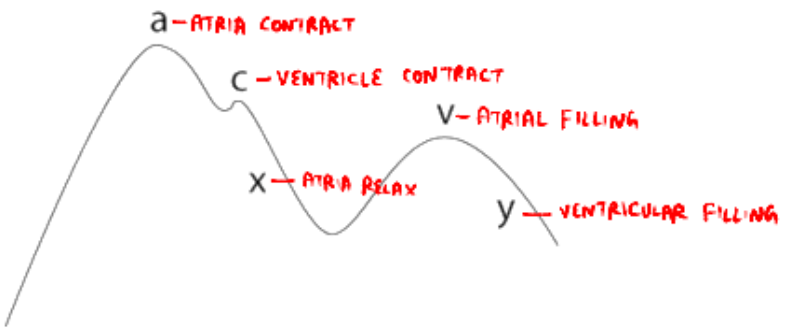
S1 - AV-Valves Close

S2 - SL - Valves Close

S3 - Early Diastole/ Rapid ventricular filling

S4 - Late Diastole/ Late ventricular filling

- JVP - Juglar Vein Pressure



- Volume

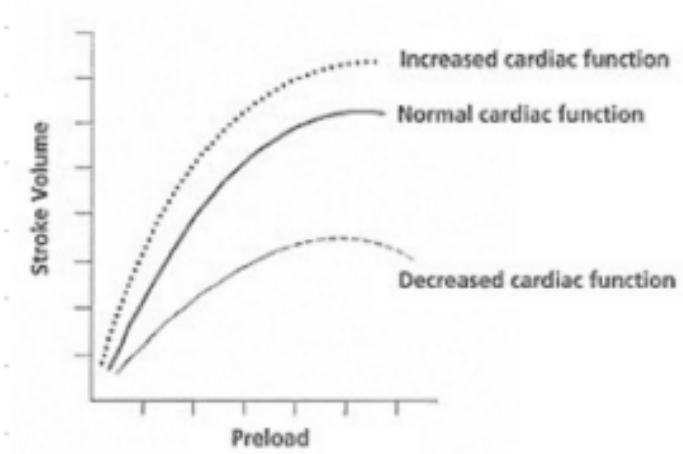
  1. End Diastole Volume → 120-130ml
  2. Stroke Volume → 70-80ml
  3. Ejection Fraction → (SV/EDV) *100 = 65%
  4. End Systolic Volume → EDV - SV = 40-50ml

- Co - Cardiac Output

  → Blood pumped by each ventricle per minute

  → SV X HR = 70x72 = 5 l/min

- Frank Starling Curve



- Coronary Blood Flow = 250 ml/min

  ↳ Max in Diastole

  ↳ Min in Systole

  - Max cross sectional area = Capillaries

- Type of Vessel

  1. Wind Kessel Vessel

     → Large elastic vessel has elastic recoil

Example - Aorta

2. Capacitance Vessel

→ They stretch & store blood

Example - Systemic Veins [Store - 50-60% of blood]

3. Exchange Vessel

→ Capillaries

4. Resistance Vessel

→ Muscular arteries & arterioles

- Flow Pressure Relationship

  Flow = Pressure/Resistance

- 4th power effect for radius

  ↳ If We ↑ radius by 2 times

  ↳ Resistance will 2^4 times = 16times

  - Pulse Pressure

    ↳ SBP - DBP = 120-80 = 40 mmHg

  - Mean Bp

    ↳ 2/3 DBP + 1/3 SBP = 95-100 mmHg

  - CVS regulatory centre's +nt in medulla

Medulla

Vaso Motor Centre [VMC]

↑HR, CO, BP

Cardio Vagal Centre [CVC]

↓HR, CO, BP

- BARO-Receptor's
  ↳ Location
    1. Aortic sinus BR → Arch of Aorta
    2. Carotid sinus BR → Bifurcation of common carotid artery
  ↳ Function → They act as break's
    +ve CVC & -ve VMC

GIT

- 4 layer's of GIT -
  1. Mucosa
     ↳ Villi & Microvilli +nt → Area of absorption
  2. Submucosa
     ↳ Contain blood vessel, nerve, lymphatic, glands
  3. Muscularis
     ↳ Muscle contract & provide movement [Peristalsis]
  4. Serosa
     ↳ For attachment & stability
     ↳ -nt in Esophagus
- GIT functions are controlled by enteric nervous system
  ↳ Parasympathetic - Receptor - M1, M2, M3, M4, M5
  ↳ Sympathetic - Receptor α1, α2, ß1, ß2
- Pacemaker of GIT - Intestinal cells of Cajal
- Cajal ⟵ Food is stimulus → BER [Resting Current]

[Active Current] Spike = ↑Ca^{2+}

↓ Calmoduline

Activate MLCK → Break ATP

→ Contraction

- Types of GIT movements

1. Peristalsis

- Move bolus forward in GIT

Relax/ open GIT



↳ Contract/ cause GIT

2. Reverse Peristalsis

Ex - Vomiting

3. MMC [Migratory Motor Complex]

or

[Hunger Pangs]

→ They are produced by empty stomach

→ k/as broom stick of GIT

→ Prepare GIT for next meal

→ Duration = 90-100 minutes

4. Mass Movement

→ For Defication

→ Normal = 3 times/ day

→ >3 Times - Diarrhoea

→ Pacemaker - Ganglionic cell of colon [Auerbach Plexus]

- Max K+ conc. → Colon
- Max K+ secretion → Saliva
- Max git secretion → Intestine - 2-2.2 l/day
- Min git secretion → Bile - 0.5-0.8 l/day
- Most acidic secretion → Gastric - 1.5-2 pH
- Most alkaline secretion → Pancreatic - 8 pH
- Brunner gland is +nt in Duodenum
- Parietal/ Oxyntic cell [Max at fundus]

  ↳ HCL

- Chief/ Zymogen Cells

  ↳ Pepsinogen $\xrightarrow{\text{HCL}}$ Pepsin

- Mechanism of HCL secretion

$CO_2 + H_2O \rightleftharpoons H_2CO_3$ [Carbonic acid]

H^+ ; $HCO3 / Cl^-$ [Exchange]

HCL

- Most Potent Choleretic→ Bile salt > Bile acid

General Physiology

- Cell Membrane Has

  ↳ 40-45% Lipid & 50-55% Proteins

- Polar substances are water soluble
- Non-polar substance are fat soluble
- Universal solvant → Water
- Ion's Na+, Cl^-, K^+, Ca^{2+} can't cross lipid CM

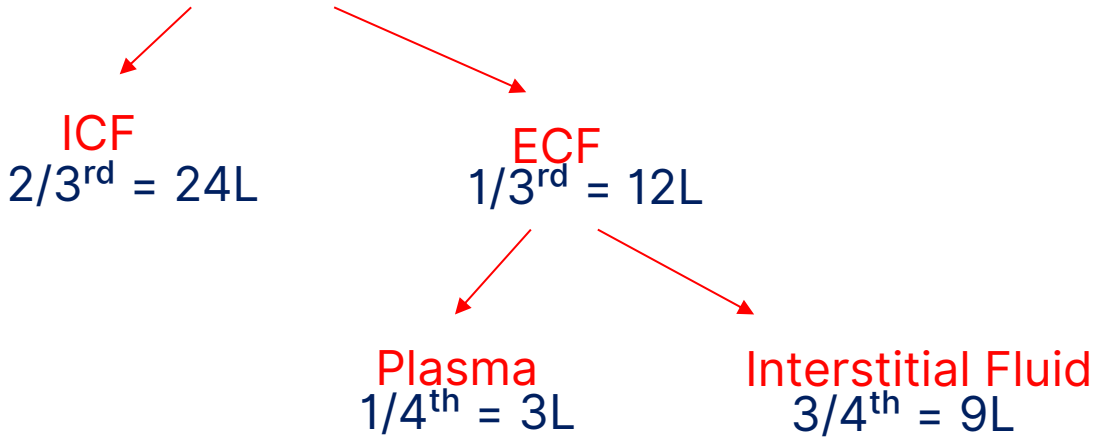
  ↓

  So, they use protein as Ion Channel
- Water use Aqua Porins for transport
- Water soluble solutes Ex. Sugar, Amino Acid etc

  They require glut

GLUT1	• Blood • Blood-Brain Barrier • Heart (lesser extent)
GLUT2	• Liver • Pancreas • Small Intestine
GLUT3	• Brain • Neurons • Sperm
GLUT4	• Skeletal Muscle • Adipose Tissue • Heart
GLUT5	• Enterocyte of Intestinal Epithelium (Luminal Side)

- Total body water is 60% = 36 L of 60L



- Dye dolution method for volume estimation

→ TBW = D_2O, T_2O

→ ECF = Inulin, Sucrose

→ Plasma = T^{125}, T^{131}

→ ICF = TBW - ECF

- Active Transport
  - → Against gradient
  - → Use A TP
  - → Eg Pump [Na^+, K^+, ATPase]

- Passive Transport
  - → Along Gradient
  - → No energy required
  - → Example - Osmosis [water]
    Diffusion [solutes & gases]

- 1° Active Transport
  - ↳ Direct use of energy
  - ↳ Example Pump
- 2° Active transport
  - ↳ Indirect use of energy
  - ↳ Example - Na^+ glucose symport
    Na^+ AA symport

PATHOLOGY

Cell Adaptation

1. Hypertrophy =
   - ↑ Cell size

   Example → Physiological
   - Uterus during pregnancy
   - Breast during lactation

   → Pathological . LVH

2. Hyperplasia =
   - ↑ Cell number

   Example → Physiological . Breast during preg. & puberty

   → Pathological . Ab ↑ in hormones

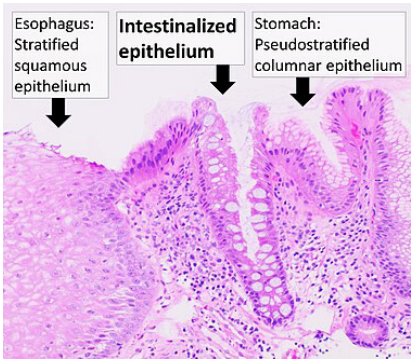
   ↳ ↑ Androgen → BPH

   ↳ ↑ Estrogen → Endometrial Ca.

3. Atrophy =
   - ↓ Cell size & number

4. Metaplasia =
   - N cell is replaced by another N cell

   Example → Squamous Metaplasia - Seen in smokers



Columnar Metaplasia → Barrets esophagus
→ Sq replaced by columnar cells
→ Hallmark = Goblet cell
→ Stain Used = Alcian Blue/Mucicarmine

Cell Injury

1. Reversible Cell Injury
   - Earliest Ion Involved → Na^{2+}
   - Most Imp. Morphological Feature → Swelling
   - Only pyknosis is seen

2. Irreversible Cell Injury
   - Earliest ion involved → Ca^{2+}
   - Mylein figures seen
   - Large amorphous density seen
   - Pyknosis

     Karyorrhexis Seen on light microscopy

     Karyolysis

3. Oxidative Stress
   - ↑Free radical → Oxidative stress
   - Most Potent Free Radical → Hydroxyl Radical
   - Consequence of free radical injury =
     - ↳ Protein misfolding
     - ↳ Damage to DNA

- Brain is protected from free radical injury

  By SOD-1 [Supera Oxide Dismutase]

Cell Death

1. Necrosis → Always pathological

   → Inf. +nt

   → Denaturation of protean & enzymatic digestion of cell's seen

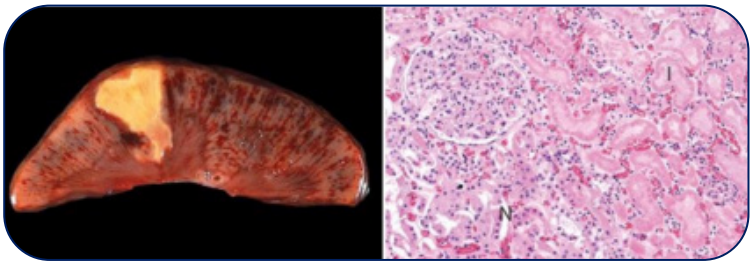
Types

A) Coagulative Necrosis

→ Mc Type of necrosis

→ Occurs in solid organs → Heart, Spleen, Kidney etc except brain

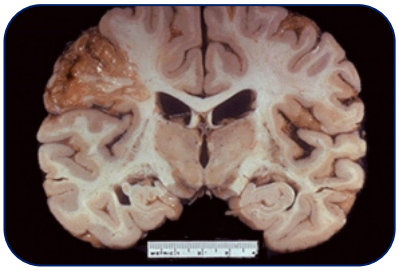
→ Considered as dry gangrene



B) Liquefactive Necrosis

→ Seen in brain

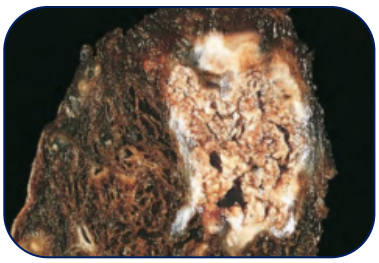
→ Wet Gangrene



C) Caseous Necrosis

→ Seen in TB, Syphilis, Histoplasmosis

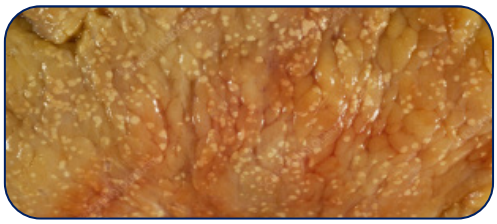
→ Cheese like app. is seen



D) Fat Necrosis

→ Chalky white deposits are seen

→ Seen in pancreas, omentum, mesentery, breast

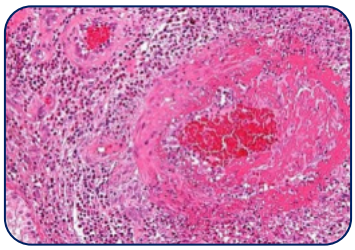


E) Fibrinoid Necrosis

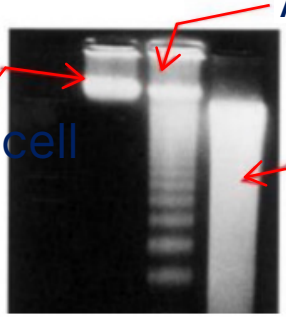
→ Seen in type 11/111 hypersensitivity rxn.

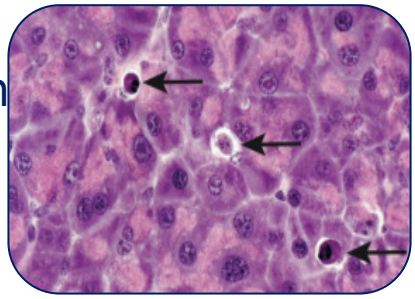
→ Seen in vessel wall

→ Pink fibrin like app. seen.



Dx of necrosis =





2. Apoptosis → Programmed cell death
→ Can be pathological or physiological
→ Earliest Feature → Cell shrinks
→ Most characteristic feature → Chromatin condensation
→ Inflammation -nt
→ Apoptotic bodies are seen
→ Enzymes involved in apoptosis

1. Caspases → Initiator - 8, 9,10
Executioner - 3, 6, 7

2. Endonuclease → DNA Breakdown → Step Ladder Pattern

→ Factors in poptosis

1. Pro apoptotic factor → BAX, BAK
2. Anti-Apoptotic → BCL-2, BCL-XL, MCL-1
3. Sensors → BIM, BID, BAD, NOXA, PUMA

3. Necroptosis

→ Necrosis + Apoptosis
→ Caspase independent

4. Pyroptosis

→ Fever + Inflammation

- Calcification =

Dystrophic	Metastatic
• Seen Dead Tissue • Serum Ca^{2+} normal	• Seen In Living • Serum Ca^{2+}↑

- Pigments =

Exogenous	Endogenous
→ Tattoo	→ Melanin → Hemosiderin → Lipofuscin aka → wear tear pigment

- Inflammation =

Acute	Chronic
• Short Duration • Neutrophil ← cells →	• Long duration • Monocytes Lymphocyte Macrophage

- Cardinal sign of inflammation
  - Rubor -Redness
  - Calor - Heat
  - Dolor - Pain
  - Tumor - Swelling

  by - Celsus
  - Functiolaesa by- Virchow

Acute Inflammation

Vascular Events

↓

Vasodilation

↓

Avascular Permeability

↓

Stasis

↓

Migration

Cellular Events

↓

Roling E-SELECTIN
P-SELECTIN

↓

Adhesion I-CAM
V-CAM

↓

Transmigration PECAM-1D
CD 31

↓

Chemotaxis C5A . LTB4 . IL-8

↓

Opsonisation

↓

- C3B, C4B, C5B
- TGG

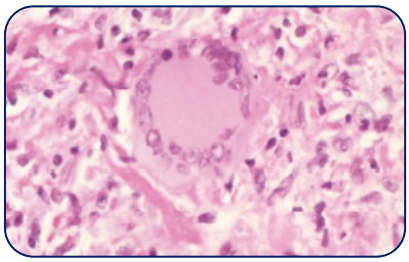
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Phagocytosis

Giant Cells

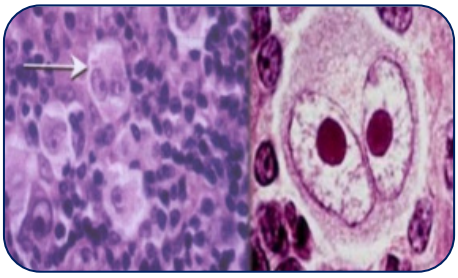
1. Langhans Giant Cells

→ Seen in - TB



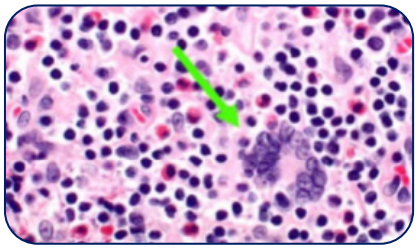
2. Aschoff Giant Cell

→ Rheumatic Heart Disease



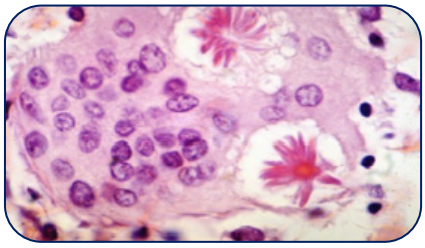
3. Reed Sternberq Cell

→ Hodgkin Lymphoma



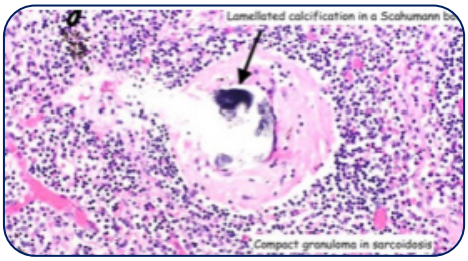
4. Warthin Finkedly Giant Cell

→ Measles



5. Asteroid Bodies

→ Sarcoidosis



6. Schauman Bodies

→ Sarcoidosis

- Histamine
- PAF [Platelet activating factory]

→ Vasodilation
→ Broncho

- Serotonin
- PGF2

→ Vasoconstriction

- Nitric Oxide
- PGD2

→ Vasodilation

- TXA_2
  - Vaso-Constriction
  - ↑Platelet Aggregation
- PGI_2
  - Vasodilation
  - ↓Platelet Aggregation
- PGE_2
  - Fever
  - Pain
- Strongest/ Potent B.C = LTC4
- MAC = C5, 6,7, 8,9

• Hypertrophic Scar → Colour same as skin → Location = Anywhere 	• Keliods → Colour darker than skin → Location - Pinna Sternum Shoulder 

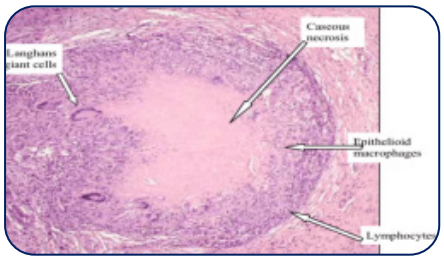
Granuloma =

- Caseating Granuloma

→ TB

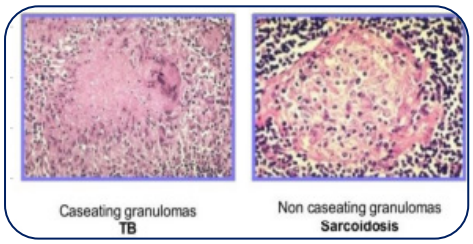
→ Syphilis

→ Histoplasmosis



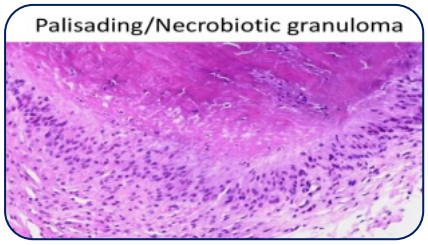
- Non Caseating Granuloma

→ Sarcoidosis



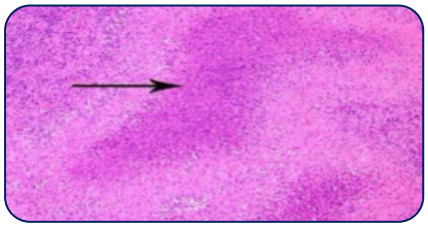
- Palisading Granuloma

→ Wegener's Granuloma



- Stellate Granuloma

→ Cat Scratch Disease



Hypersensitivity =

1. Type- I = Allergy
   Example - Asthma, Allergy, Anaphylaxis
2. Type- II = Blood Related
   Example - My Mysthenia Gravis
   Blood Blood Transfusion
   Group Good Pasture Syndrome
   Is Immune Thrombocytopenic Purpura
   Rh Rh Incompatibility
   Positive Pernicious Anemia
3. Type III = Immune Complex Mediated
   Example = S → SLE
   H → Henoch Schelin Purpura
   A → Arthus Rxn
   R → Rheumatoid Arthritis
   P → Poly Arthritis Nodosa
   → Type II Lepra Reaction
4. Type IV = Delayed type
   → Type I Lepra Rxn

- Types of graft
  Auto → Self graft
  Iso→ Identical twins
  Allo → Same species
  Xeno → Different species
  → Hyper Acute Graft Rejection → Min - Hrs

→ Acute graft rejection → day's - weeks

→ Chronic graft rejection → month - year's

- β- Cells are produced in BM & mature in BM
- T-Cells are produced in BM & mature in Thymus

- MHC- I = Present Ag to CD8 T-cells
- MHC- II = Present Ag to CD4 T-cells

Types of Cells

Permanent	Stable	Labile
• Non-dividing • Left cell cycle	• Divide when need • G_o phase	• Rapidly dividing • G_1 phase

Repair =

- Deposition/ formation of Granulation Tissue
- Hallmark of Granulation Neovascularization

Wound Healing =

1° Intention		2° Intention
• Wound is clean • No loss of tissue		• Wound is infected • Tissue loss +nt

- Imp. Steps

O Hrs - Blood Clot

0-24 Hrs - Neutrophils

Day 3 - Neutrophils ⟶ Macrophage + Granulation Tissue Appears

Day 5 - ↑ Granulation

↑ Neovascularisation

Day 14 - Maximum Collagen

Day 28 - Scar formation

- Carcinoma → Blood
- Sarcoma → Lymphatic

- **Neoplasia =**

Benign	Malignant
→ Suffix = OMA → Anaplasia → -nt → Slowly Growing → Metastasis - nt → Local invasion - nt	→ Suffix = Carcinoma Sarcoma → Anaplasia → +nt → Rapidly growing → Metastasis +nt → Local invasion +nt

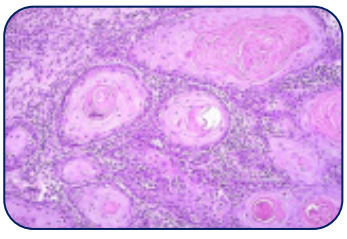
Exception = Malagnancy that end with OMA

- Melanoma
- Lymphoma
- Chloroma
- Seminoma

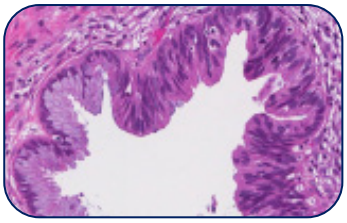
Exception = Malagnancy that do not metastasize

- Basal Cell Carcinoma
- Glioma

- Pointer to Dx cancer →



Keratin Pearls
↳ Sq. cell carcinoma



Abnormal Glands
↳ Adenocarcinoma

- Cell cycle

- Most radiosensitive phase G_2M
- Most radio resistant phase S
- Most radiosensitive cell Lymphocyte
- Most radio resistant cell Platelet

- Tumour Suppressor Genes [TSG]

TSG	Chromosome	Tumour
Rb	13q/14	• Retinoblastoma
BRCA-1	17	• Breast Cancer • Ovarian Cancer
BRCA-2	13	• Breast Cancer
NF1	17	• Prostate Cancer • Neurofibromas
NF2	22	• Schwannoma
WT-1 & WT-2	11	• Wilm's Tumour
VHL	3	• Clear cell Rcc
STK 11		• Peutz Jegher syn

- Imp. Tumour Marker's

• PSA	• Prostate cancer
• Calcitonin	• Medullary carcinoma of thyroid
• HCG	• Choriocarcinoma
• AFP	• Hepato cellular carcinoma
• Ca 19-9	• Pancreatic cancer
• Ca 125	• Ovarian cancer
• Ca 15-3	• Breast cancer
• Catecholamines	• Pheochromocytoma

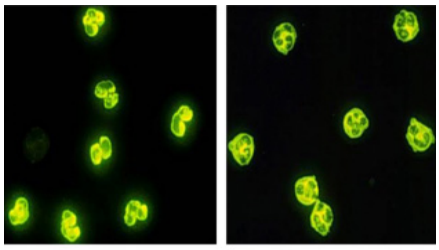
- **Vasculitis**

I Large Vessel Vasculitis
1. Temporal arteritis / Giant cell arteritis
2. Takayasu arteritis

II Medium Vessel Vasculitis
1. Pan [Poly arteritis nodosa]
2. Kawasaki Disease
3. Buerger's Disease

III Small Vessel Vasculitis
1. MPA [Microscopic Polyangiitis]
2. Wegener 'S Granulomatosis
3. CSS [Chrug Straus Syndrome]
4. HSH [Henoch Scholein Purpura]

Paras ki
Kawa Saki
Bike
My
way to
C
H
Chandigarh


P-ANCA Pattern C-ANCA Pattern

P - ANCA +nt in
C- Churg Strauss Syndrome
M - Microscopic Polyangiitis
C - ANCA +nt in
Wecener's Granulomatosis

- Dx - of Vasculitis - Ab/ Biopsy
- Rx of all Vasculitis - Steroid

Except 1. Buerger's Disease → Stop Smoking
2. Kawasaki Disease → I.V -Ig
3. Wegner's Granulomatosis → Cyclophosphamine

- Temporal Arteritis/ Giant Cell Arteritis

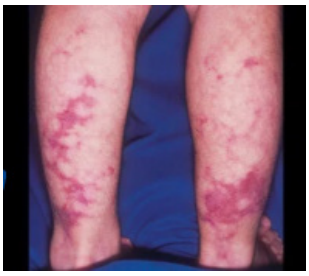


C/F
  - Facial pain
  - Pain in chewing/ mastication
  - Head ache
  - Polymyalgia

Dx Temporal Artery Biopsy

Rx Steroids → By rule of 60
  - Female > 60years
  - ESR >60
  - Rx 60mg Prednisolone

- PAN (Poly Arteritis Nodosa)



  - Condition occurs due to inflammation
  - Type III HSN
  - Involve multiple organs except lung
  - Dx - Biopsy
  - Rx - Steroid

- Kawasaki Disease



Rx PCM

IL-1 Antagonist - Anakinra

Toc → IV Ig

- Chrng Strauss Syndrome =
    - Associated with asthma/ eosinophilia
    - IgE may ↑



Dx - P-ANCA

Rx - Steroids

- Henoch Schonlein Purpura
    - Mc Vasculitis in children

Associated τ IGA nephropathy

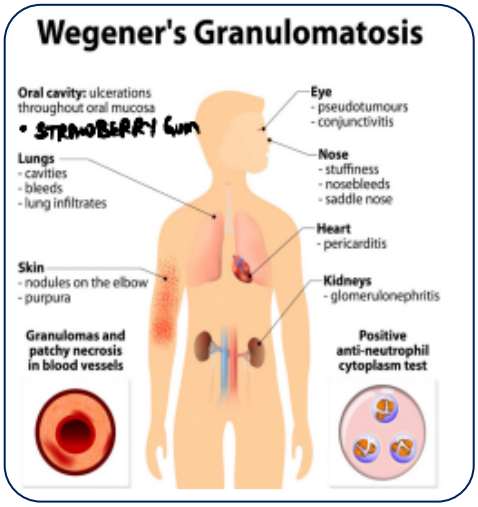
C/F Palpable purple rash +nt

Haematuria

Dx Clinical Symptoms

Rx Self Limiting (90%)

Low Dose Steroid x 7 days



- Weceners Granulomatosis

Dx - C - ANCA

Doc - Cyclophosphamide

- Buergers Disease/ Thrombo Angitis Oblitterans
    - H/O Young Male Smoker
    - Artery → Vein → Nerve

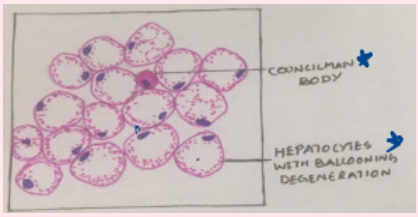
Never Lymphatics



Rx - Stop smoking

- Amputation

- Liver
  → Mc Type of hepatitis - Hep-A
  → Mc Type of hepatitis in children - Hep-A
  → Mc Type of chronic hepatitis - Hep-B
  → Most potential cause of chronic hepatitis - Hep-C
  → Mc Type of acute hepatitis in adult - Hep-E
  → Mc Hepatitis associated τ preg. - Hep-E

Acute Viral Hepatitis	Chronic Viral Hepatitis
< 6month H/P 	>6 Month H/P Bridging Fibrosis Ground Glass Hepatocyte

- Marker's of Hep.B
  $HBsAg$ Virus entry
  $HBcA9$ Not detected
  $HBeAg$ Virus replication = Infectivity +ve
  Anti HbcAb ⟨ IgM Acute
  IgG - Chronic
  Anti HbeAb - End of infectivity
  Anti HbsAb - Cured

  Rx of Hep. B → Tenofovir
  Entecavir
  Lamivudine
  Rx of Hep. C → Ledipasvir

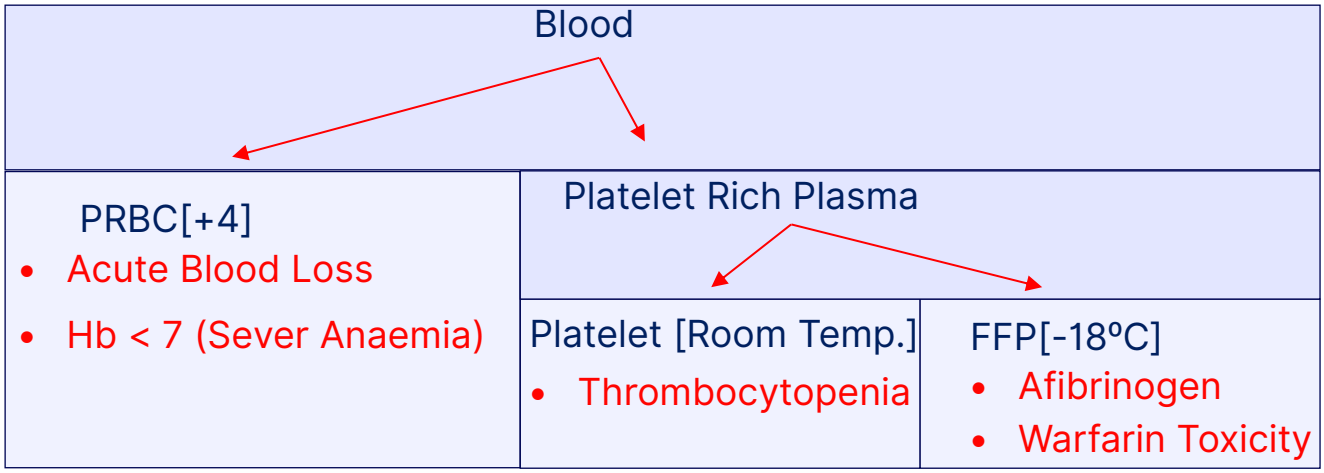
Sofosbuvir

- **Biliary Tract Disorder**
  1) 1° Biliary Cirrhosis
     - AMA +ve [Anti - Mitochondrial Ab.]
     - Rx - Ursodeoxycholic acid
     - TOC - Liver transplant
  2) 2° Biliary Cirrhosis
     - Cause - Obstruction
     - Rx - Treat the cause
  3) 1° Sclerosing Cholangitis
     - P-ANCA +ve
     - Rx - Ursodeoxycholic acid
     - TOC- Liver transplant

- **Haematology**
  - Mc Type of blood group - ABO
  - Least common type - Bombay Phenotype
  - Safest blood for transfusion - Cross Matched
  - Universal donor - O
  - Universal recipient - AB

- **Pretransfusion Testing**

  HIV, Hep B, Hep C, Syphilis, Malariae



- BT notes =
  - Warming prior τ body heat
  - 6-8 drops/ min
  - ↑Rate by 10-14 drops/ min
  - In case of any hypersensitivity stop B.T

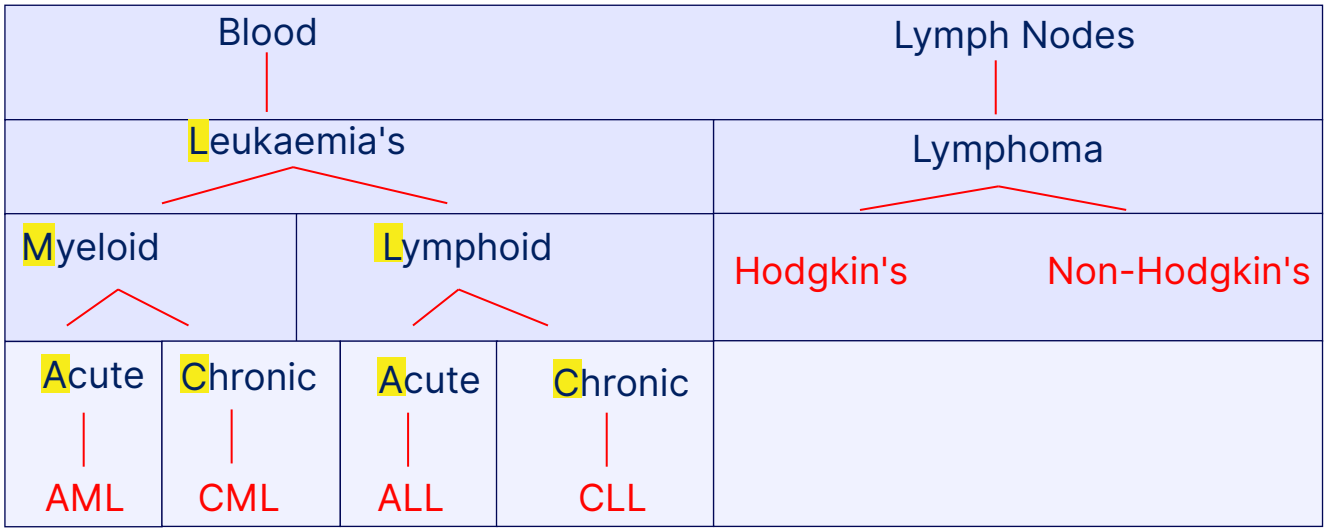
    Rx Inj. Avil

    Inj. Dexamethasone
  - In case of hypotension

Rx Epinephrine

- **WBC Disorders =**

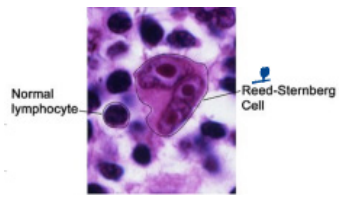
Neoplastic Cells



- **Hodgkin's Lymphoma**

Etiology - Ebstein Barr virus associated
↓
Produce latent membrane protein-1
↓ [activate NF -Kb pathway]
↑ Cell Proliferation
↓
Hodgkin's Lymphoma

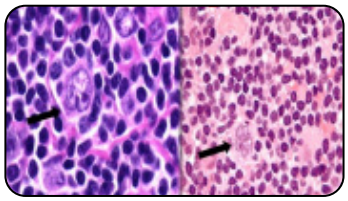
C/F :
- PEL Ebstein fever, Night sweat, Weight loss
- Painless cervical lymph node enlargement
- Symptom worsens with alcohol

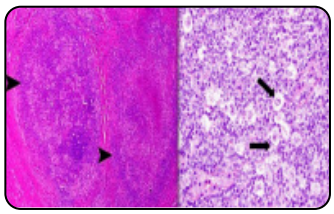


H/P-

Lacunar variant
- Empty space around
- Seen in nodular sclerosing H.L

Popcorn variant
- Seen in (NLPHL) Nodular lymphocyte predominant H.L





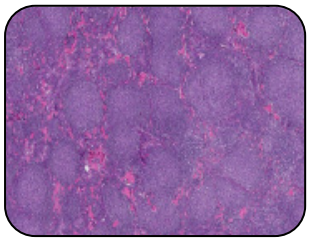
- Types
  - Lymphocyte Rich
  - Lymphocyte Poor Worst prog.
  - Mixed Cellularity Mc type India
  - Nodular Sclerosis Mc in world
  - Lymphocyte Predominant Best Prog.

  Rx- A - Adriamycin
  B - Bleomycin
  V - Vinblastine
  D - Dacarbazine
- Non-Hodgkin's Lymphoma

  β - cell associated

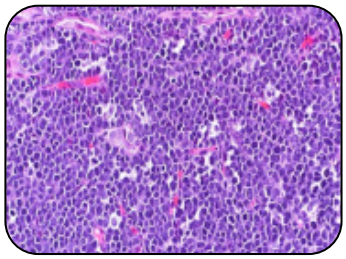
  Tumour marker - CD19, CD20

  1. Follicular Cell Lymphoma →
     - ß- cell lymphoma
     - t(14, 18)
     - BCL-2 gene overexpression



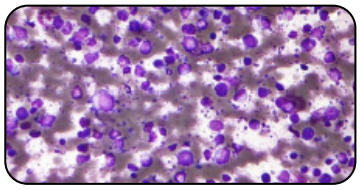
→ Cells are arranged in follicles

  2. Burkitt Lymphoma
     - t(8,14)
     - C-Myc gene overexpression

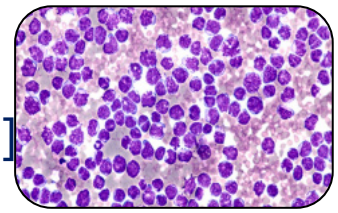


- On lymph node biopsy → Starry sky app.

C/P . Jaw swelling +nt



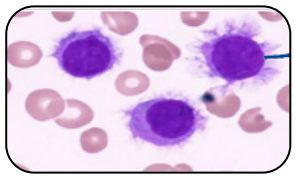
3. DLBCL - Diffuse large ß cell lymphoma
   - Mc Type & dangerous type with worst prognosis
   - Myc - Translocation



4. Mantle Cell Lymphoma
   - Seen in eldervly [Male > Female]
   - t(11,14)
   - ↑ Expression of Cyclin D1 Gene

Lymph Node Biops

Small Cleared Lymphocytes



5. Hairy Cell Lymphoma

Hairy Cell

Dx - Trap +ve [spl. stain]

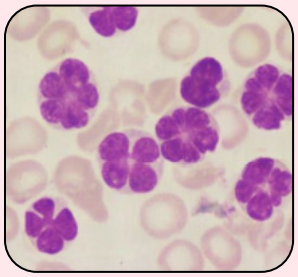
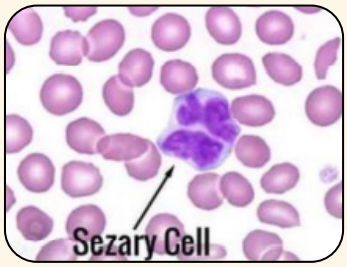
[Tartarate Resistant Acid Phosphatase]

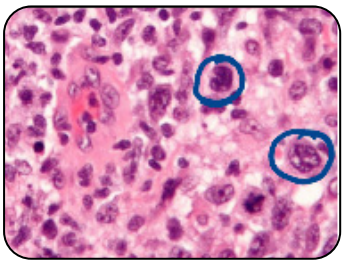
- Hb, TLC, Platelets - ↓

Bm-Biopsy - Honeycomb/ Fried egg appearance

Rx - Cladribine

- T-cell associated

1. Adult T-Cell Leukaemia	2. Mycosis Fungoides/ Sezary Syndrome
→ By HTLV-1 [Human T-Cell leukaemia virus]  → Flower cells	Not a fungal infection 



3. Anaplastic Large T-Cell Lymphoma

Pathogenesis - ALK Gene mutation on chromoso Zp

H/P - Hallmark Cells

- Imp - Classification for lymphoma - Real classification
- Staging Method - ANN Aarbor method
- Rx of non Hodgkin's lymphoma

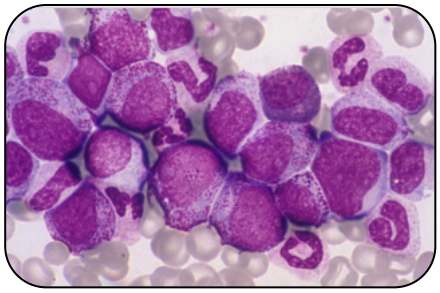
R - Rituximab

C- Cyclophosphamide

H - Hydroxyduano Rubian

O - Oncovin (Vincristine)

P - Prednisolone



- **Myeloid Disorder's**

1. CML
    - C/F → Hepatosplenomegaly
    - (9:22) : Philadelphia Chromosome

↓

ABL-BCR gene active

↓

Tyrosine Kinase - Active - To Block - Doc - IMATINIB

↓

Cancer

Toc - BMT

Lab Dx-
- Hb - normal
- TLC ↑↑↑
- Platelet count ↑↑↑
- Myeloid lineage cell ↑↑↑

Bm. Biopsy → Garden party appearance seen

2. AML
    - Intracytoplasmic rods → Auer Rods
    - Dye - MPO [Myelo Peroxidase]

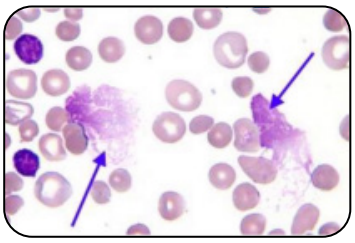
Sudan Black

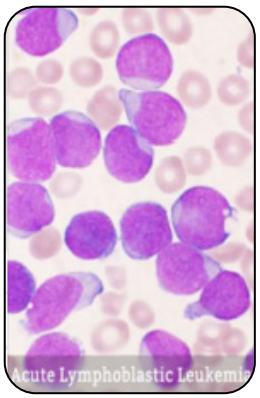
- Types - Mc Type - M_2
  Most Significant M_3 → t(15;17)
  Gingwal Infiltrate M_5

Rx - BMT

3. CLL

- CLL → Richter transformation → DLBCL (Most severe)
- Mc adult/ elderly
- A/W → AIHA [Auto immune haemolytic anaemia]
- HPE: Smudge/ Parachute/ Basket Cell
- Doc - Fludrabine
- Periodic Follow-up - 1-2 yr - Pet scan



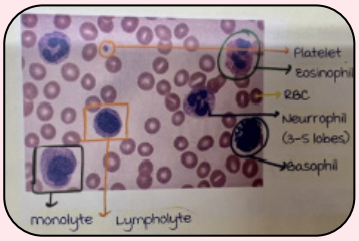


4. ALL

- Mc in children
- L1 - MC
- L3 - Worst prognosis
- A/W Down syndrome

Rx V - Vincristine
A - Asparaginase
P - Prednisone
O - Doxorubicin
TxOC - Bone Marrow Transplant

- Haematopoiesis =
  - 3"d Week of intrauterine life Yolk Sac
  - 3"ª Month of intrauterine life Liver
  - 4th Month of intrauterine life B.M & Liver
  - Just before birth Bone Marrow
- Hematopoietic Stem Cell

Lymphoblast	Myeloblast
• T-Lymphocytes • B-Lymphocytes • nk - cells 	• Erythroblast → RBC • Megakaryoblast → Platelet • Myeloblast → Neutrophils, Eosinophils, Basophils • Monoblast → Monocyte → Macrophage

- **Iron deficiency anaemia**
  - Mc type of anaemia
  - Iron in blood = Transferrin
  - Iron is stored in body = Ferritin
  - Iron is absorbed in duodenum via DMT-

Dx - Hb ↓
MCV ↓
Serum Iron ↓
Serum Fermin ↓

TIBC - Total iron binding capacity
FEP - Free Erythro protoporphyrin

Cause - Dietry Def
Excess Blood Loss - Trauma Etc.
P. Smear → Microcytic Hypochromia Cells
Pencil Shape Cells
Special Feature → Koilonychia
Cheilosis
Rx → If Hb < 7 = RRBC
If Hb < 7 = Oral -> Ferrous Sulphate
I.V -> Ferrus Sucrose/ Carboxymaltose

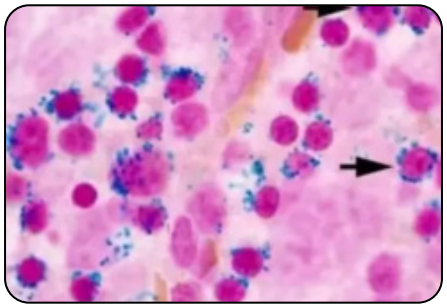
- Sideroblastic Anaemia -

  Dx - S. Iron ↑
  S. Ferritin ↑
  MCV ↓
  TIABC ↓

  P. Smear →

  - Basophilic Stipling

  Cause -
  - Lead Poisoning
  - Iron overload



  Rx BMT

- AOCD - Anaemia of Chronic Ds -
  - Chronic Inf.
  - Macrophage → Hepcidin → Lock iron within stores

  Dx → S. Fermin ↑
  S. Iron ↓
  MCV ↓
  TIBC ↓

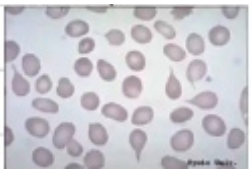
  Rx → Treat the cause

- Thalassemia =
  - β-Thalassemia
  - β- chain defect

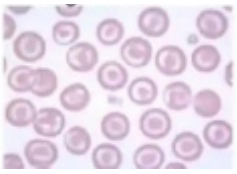
- Types
  - Minor - No symptom - No. Rx needed
  - Intermediate - Mild to mod. symp - B.T every 3-6 month
  - Major - Severe symp. - BT every month
- Dx -
  - MCV ↓
  - MCV/ RBC (Mentzer Index) = <13

- P. Smear =

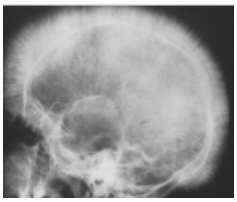
Tear Drop Cell



Target Cell



- C/F



Crew cut app



Chipmunk face

- SCA - Sickle Cell Anaemia

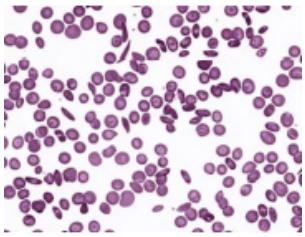
Glutamic Acid - Valine at 6th position of β-globin chain

C/F =



- Jaundice
- Gallstone
- Bone Pain
- Auto splenectomy

P. Smear



Sickle Cell RBC

Dx - MCV-N
Hb ↓
Rx BT

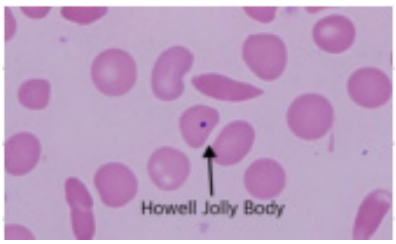
- Splenectomy
  - Auto splenectomy → Sickle cell anaemia
  - Planned → H. Spherocytosis

RBC spherical

→ H. Elliptocytosis Ape like

→ AiHA (Auto immune Haemolytic Anaemia)

Post Splenectomy → Howell Jolly bodies seen



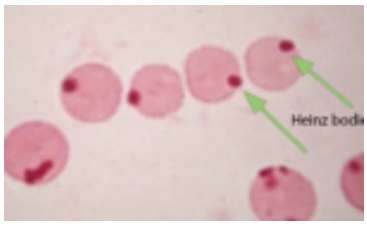
- G_6PD Deficiency Anaemia

  XLR

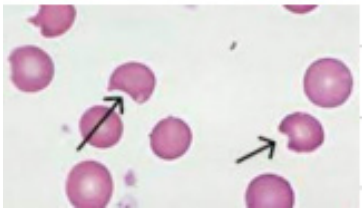
  G_6PD def → RBC at oxidative damage

  C/F
  - No sym. at birth/ rest
  - Stress ↑ symptoms

  Dx-
  - MCV - normal



  - P.Smear →
    - Heinz bodies →
    - Bite cells →



- AiHA - Autoimmune haemolytic anaemia
  - Associated CLL

  Dx - Coomb Test

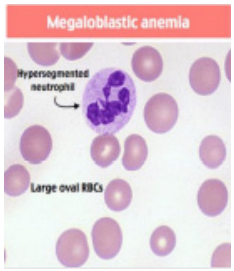
  Rx - Steroid

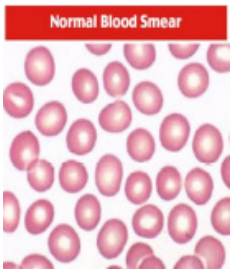
  ↓

  Iv Iq

  ↓

  Splenectomy
- Megaloblastic anemia





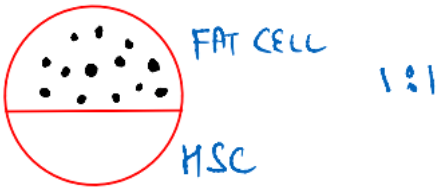
Cause

- MCC - Vit B12 deficiency
- Pernicious anaemia
- Pancreatitis
- IMMERSLAND GRESBOCK Synd.

Megaloblastic anaemia

- Folic Acid deficiency
  >100
  +nt
  FIGLU Test
  MCV
  neurological defect
  test
- B-12 Deficiency
- anaemia
  >100
  -nt
  Shilling Test

- Aplastic anaemia
  - Aplastic anaemia -> HSC lost
  - B.M Biopsy → Only fat cell
  - Dx - RBC ↓
    WBC ↓
    Platelets ↓



Rx - • BMT
• BT
↑ RBC – Darbopoitin
Epoitin
↑ Platelet Eltrompopag
↑WBC Filgrastin & Sargramustin

PHARMACOLOGY

→ Pharmacokinetics = What body do to a drug

→ Pharmacodynamics = What drug do to a body

- Pharmacokinetics

A → Absorption - From administration site to blood
D → Distribution - Amount of drug going to tissue
M → Metabolism - Make drug water soluble
E → Excretion - Coming out in urine

A. Absorption

→ Movement of drug from site of administration to blood

→ Absorption depends on lipid solubility

↳ Bioavailability = B.A

→ % of given dose which reach systemic circulation

- Bioavailability $\propto \frac{1}{\text{1st pass metabolism}}$
- B.A & Absorption

Example - I.V route have 100% bioavailability

B. Distribution

→ Amount of drug going to the tissue

↳ Volume of distribution (Vd) = $\frac{\text{Amount Given}}{\text{Piasma con. of drug}}$

↳ Loading Dose (LD) = Vd X TPC

↳ Maintenance Dose (MD) = Clearance X TPC

↳ Vd $\propto \frac{1}{\text{Plasma Protein Binding}}$

↳ Vd o Lipid Solubility

C. Metabolism

Make Drug Water Soluble

Phase I [Breking]

Hydrolysis
Cyclization
Oxidation
Reduction
Decyclization

Phase II [Conjugation]

Molecule attach to drug like

G - Glucuronide
G - Glutathione
A - Acetyl
M - Methyl

Enzyme Inducing Drugs

G → Griseofulvin
P → Phenytoin
R → Rifampicin
5 → Smoking
Cell → Carbamazepine
Phone → Phenobarbitone

Enzyme Inhibiting Drugs

Vit → Valproate
K → Ketoconazole
CAN → Cimetidine
Cause → Ciprofloxacin
Enzyme → Erythromycin
Inhibition → Isoniazid

D. Excretion

→Water soluble drugs are excreted

→Lipid soluble are reabsorbed

↳ Order of kinetics

1st Order

→Constant fraction is removed

Example = 50%

100
↓
50
↓
25
↓
12.5

Zero Order

→Constant amount is removed

Example = 20

100→80→60→40→20→Zero

Zero

W - Warfarin

A - Alcohol/ Aspirin

T - Theophylline

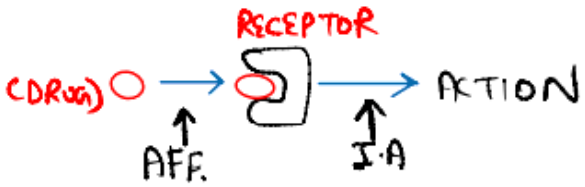
T - Tolbutamide

Power - Phenytoin

- **Pharmacodynamics = Drug to body**

↳Enzyme Inhibition

	Km	Vmax
Competitive	↑	Same
Non - Competitive	Same	↓



→ Affinity - Ability of drug to bind to a receptor

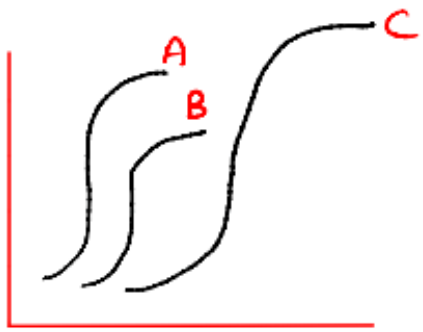
→ Intrinsic Activity - Ability to produce action after binding to receptor

→ On the basis of intrinsic activity, we have

↳ Agonist

↳ Antagonist

- Dose Response Curve



Potency = A > B > C

Efficacy = C> A > B

- Drugs that are metabolized by acetylation

S - Sulphonamide
H - Hydralazine
I - Isoniazid
P - Procainamide

→ Adverse Effect → SLE

- Clinical Trials

→ Phase 1 - Done in healthy people to know max tolerance of drug

→ Phase 2 - Done in 20-200 pt → efficacy
→ safety

→ Phase 3 - Done in 5000 pt

↳ New drug compared with old = Standard drug cont

→ Phase 4 - Post Marketing Trial

↳ Done to check rare / long term side effect

- Drug Safety

↳ Therapeutic Index = $\frac{\text{Leathal Dose}}{\text{Effective Dose}}$

↳ Narrow T.I → Unsafe drug - Need → TDM

↳ L Broad T. I → Safe drug

ANS

Parasympathetic/ Cholinergic	Sympathetic/ Adrenergic
→Major Neurotransmitter - ACH →Nerve fiber originate from - Cranio Sacral Region	Nor - Adrenaline Thoraco Lumber Region

Miosis	Eye	Mydriasis
Broncho Constriction	Bronchus	Bronchodilation
CO, BP, HR	Heart	CO, BP, HR
Diarrhoea	Git	Constipation
Flow	Bladder	Flow
Erection	Penis	Ejaculation
Secretion	Gland	Secretion

M1 - Git

M2 - Heart

M3 - All

M4 - Brain

M5 - Brain

∝1 - Blood Vessel = V.C
Eye = Dilation
Urethra = Constriction

∝2- Opposite to a1

ß1 - Heart - 1 CO, BP = V.C

ß2 - Lungs - B.D
Blood Vessel - V.D

ß3 - Urinary Bladder
- Relax Bladder

- Cholinergic Drugs
  - A Directly acting
    - ↳ Pilocarpine - Rx of ACG - M3
    - ↳ Methacholine - Act on Myocardium - M3
  - B. Indirectly acting

MOA - ACH Esterase negative

I. Lipid Soluble

→Physostigmine

Doc - Atropine Poisoning

II. Water Soluble

→Neostigmine
→Pyridostigmine — Rx Myasthenia Gravis

→Edrophonium → Dx Myathenia Gravis [Tension Test]

Imp. Atropine/ Dhatura/ Belladonna poisoning

↳ Doc - Physostigmine

Organophosphate & carbamate & early mushroom poisoning

↳ Doc - Atropine

Organophosphate alone

↳ Doc - Atropine, Oxime

- **Anti Cholinergic Drug**

According to the effect's

1. Stomach

- M1 Blocker
-Used For Peptic ULER — Pirenzepine, Telenzepine

2. Heart

-M2 Blocker

-Atropine - \ HR

3. Bronchus

-M3 Blocker

-Ipratropium
-Tiotropium
Used in bronchial asthma COPD

4. Bladder

-M3 Blocker
-Solifenacin
-Darifenacin
-Oxybutynin
Used in over active bladder

5. Eye

-M3 Blocker
-Atropine - Children
-Tropicamide - Adult
Action - Mydriasis Cycloplegia

Sympathetic Nervous System

∝1

Blood Vessel - VC
Eye - Mydriasis
Prostatic Urethra - ↓Flow

∝2

Act as Brek on ∝1

ß1

Heart - ↑ BP, CO = V.C
Kidney - Secretes Renin

ß2

Lungs - Broncho Dilation
Git - Function
Bladder - ↓ Function

Blood Vessel - V.D

ß3

Urinary Bladder - ↓ Function

- Adrenergic Drugs

I. Dopamine [Stimulate ∝1, ß1, D1]

- Doc - Shock with oliguria
MOA - ∝1 - VC to Rx shock

D1- VD in kidney to Rx Oliguria

II. Nor-Adrenaline

→ Used in shock

III. Adrenaline

→ Doc - Anaphylactic Shock & Cardiac Arrest

→ In anaphylactic shock

1:1000 - I.M Route > 1:10,000 - I.V Route

→ In cardiac arrest

1:10,000 - I.V Route

- Anti Adrenergic Drug

I. Non-selective a Blockers

→ Phenoxybenzamine - Doc - Pheochromocytoma

→ Pnentolamine - Doc - Cheese Rxn

II. Selective a1 Blockers

→ Prazosin
→ Terazosin
→ Doxazosin

Doc BPH + HTN

S.E-Postural Hxpotention

III. Selective 1A Blockers

Better than a1 negative bcz it do not cause postural hypotension

→ Tamsulosin
→ Silodosin

Doc BPH

IV. Selective ß1 Blockers

N - Nebivolol, Nadolol → Longest Acting β-Blocker

- Haematology

1. Drug Affecting Blood Flow
    - Tx A2 -ve - Aspirin
    - ADP -ve - Clopidogrel
    - Anti Fibrin/ Anticoagulant

    ↳ A. Vit K Antagonist - Warfarin

    Monitoring done by - PT/ INR

    C.I - In pregnancy

    Rx of warfarin toxicity

    ↳ Vit. K - If bleeding has not started

    ↳ FFP - If bleeding is already there

    B. Indirect Factor 11a Inhibitor

    ↳ Unfractionated Heparin

    ↳ LMWH

    →Heparin in anticoagulant of choice in preg.

    →Monitoring of heparin done by aPTT

    →Antidote - Protamine Sulphate

2. Drugs affecting cell's of blood
    - RBC↑ - Darbepoetin

      Epoetin
    - WBC↑ - Filgrastin

      Sargramostin
    - Platele↑ - I.V Oprelvekin

      Oral - Eltrambopag

GIT

- Peptic Ulcer Disease
  1. M-1 -ve
     - ↳ Pirenzepine
  2. H2 -ve
     - ↳ Cimentidine
     - ↳ Rantidine
  3. Proton Pump Inhibitor

     Pantoprazole
     Lansoprazole
     Omeprazole
     Rabeprazole
     Esomerrazole

     Doc - Peptic Ulcer Disease
     Gerd
     ZES
  4. Anti H. Pylori

     Triple Therapy Drug [PPI + 2 Anti-Microbial]
     - ↳ Clarithromycin
     - ↳ Amoxicillin
     - ↳ PP

  Anti - Emetic Drugs

  →Doc for motion sickness - Hyoscine

  →Doc for morning sickness - Doxylamine [HI -ve]

  →Doc for mountain sickness - Acetazolamide

  →Doc for chemotherapy induced vomiting - Ondansentron [SHT3 -ve]

IBD

A. Ulcerative Colitis

- Sulfasalazine
- Osalazine
- Mesalamine

B. Crohn's Disease

DOC - Steroid

If resistant to steroid

- Certolizumab
- Adalizumab
- Golimumab
- Entanercep

TNF x-ve
C. I = TB & Hep-B

CNS

- Anti - Epileptic Drug

MOA - ↑ GABA

↑ Glutamate

Ca^{2+} & Na^{2+} channel blocking

↳ Drug that release GABA

Gabapentin

Pregabalin

↳ Drug that inhibit reuptake of GABA

Tiagabine

Drug that act on GABA receptor

-Barbiturates - Phenobarbitone

-Benzodiazepines - Diazepam

Alprazolam

Lorazepam

-Ad Of Benzodiazepines - Flumazenil

↳ Drug ↑ action of glutamate

-AMPA -ve - Perampanel

-NMDA -ve - Felbamate

↳ Ca^{2+} channel blocker

-Ethosuximide Doc in absence seizure
< 2 yrs age

↳ Na^{2+} channel blocker

-Phenytoin

-Carbamazepine

↳ Valproate

Doc - GTCS

Absence seizures > 2 yrs age

Myoclonic seizures

Atonic seizures

S.E - Hepatotoxicity

Pancreatitis

PCOD

- **Bipolar Disorder**

Doc → Lithium

Doc in pregnancy → Atypical Antipsychotics

- **Opioids Antagonist**

Naloxone - Short Acting - I.V - Doc for acute cond.

Naltrexone - Lon Acting - Oral

- **Anti-Obesity Drugs**

Lorcaserine

Orlistat -Intestinal Lipase -ve

Sibutramine

Topiramate

- Drug used for smoking

  Vareniciline - Doc for smoking cessation

  Nicotine

- Anti - Alcohol drugs

  Naltrexone

  Ondansetron

  Topiramate

  Acamprosate

- Erectile dysfunction

  PDE-ve - Sildenafil Doc

  Tadalafil

BIOCHEMISRTY

Carbohydrate Metabolism

Pathway	Site	Rate Limiting Enz.
Glycolysis/ Embden Meyerhof Pathway [EMP]	Cytosol	PFK-1 - Phosphofructokinase-1
Kreb Cycle/ TCA Cycle/ Citric acid cycle	Mitochondrial Matrix	Isocitrate Dehydrogenase
Gluconeogenesis	Liver & Kidney Cytosol & Mitrochondria	Fructose 1,6- Bisphosphate
Glycogenesis	Liver & Skeletal Muscle (Cytosol)	Glycogen Synthase
Glycogenolysis	Liver & Skeletal Muscle (Cytosol)	Glycogen Phosphorylase
HMP shunt/ Pentose Phosphate pathway	Liver & Skeletal Muscle (Cytosol)	G&PD - Glucose - 6-Phosphate Dehydrogenase

Protein Metabolism

Pathway	Site	Rate Limiting Enzyme
Urea Cycle/ Ornithine Cycle	Liver C Cytosol & Mitochondria	CPS-1 -+ Carbamoyl Phosphate Synthase- 1

Lipid Metabolism

B- oxidation of fatty acid	Mitochondria	CAT-1 -+ Carnitine Acyl Transferase-1
Ketogenesis	Mitochondria	HMG COA Synthase
Fatty acid synthesis	Cytosol	Acetyl COA Carboxylase
Cholesterol synthesis	Cytosol	HMG COA Reductase

Nucleotide Metabolism

Purine Synthesis	PRPP Glutamyl Aminotransferase

- **Disease & Enzyme Deficiency**

A- Carbohydrates

- Von Gierke's disease

  →Disease in which body can't break down glycogen

  →Enzyme deficiency → Glucose-6- Phosphatase

- Pompe's disease

  →Disease in which body can't make protein due to mutation in GAA - GENE [Acid Alpha - Glucosidase]

  Enzyme deficiency → Acid Maltase

- Cori's disease

  →Structurally abnormal glycogen is accumulated that impairs body function

  Enzyme deficiency → Debranching enzyme

- Anderson's disease

  →Abnormal glycogen is produced and accumulated in body

  Enzyme deficiency → Branching enzyme

- Mc Arlde's disease

  →Enzyme → Muscle Glycogen Phosphorylase

- Her's Disease

  Enzyme Def. → Hepatic Glycogen Phosphorylase

- Essential Fructosuria

  →Fructose coming out in urine

  Enzyme deficiency → Fructokinase

- Fructose Intolerance

  →Lack of protein which is needed to breakdown fructose

  Enzyme deficiency → Aldolase B

- Galactosemia

  Enzyme deficiency → Galactose-1- Phosphatase Uridyl Transferase

B. Proteins

- Phenylketonuria

  →Mousy/ Musty Odour

  Enzyme → Phenylalanine Hydroxylase

- Alkaptonuria

  →Black urine disease

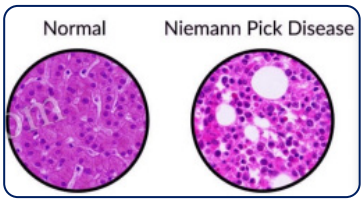
  Enzyme → Homogentisate Oxidase

- Maple Syrup Urine Disease

  →Burnt Sugar Odour

  Enzyme → Branched chain keto acid dehydrogenase/ decarboxylase

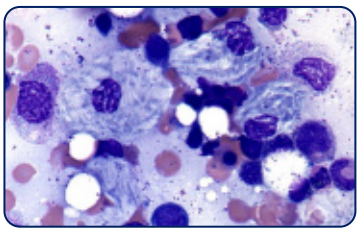
C. Lipid Storage Disease



- Niemann Pick Disease

  →Body fails to metabolic lipid & cholesterol & they get accumulated in cells

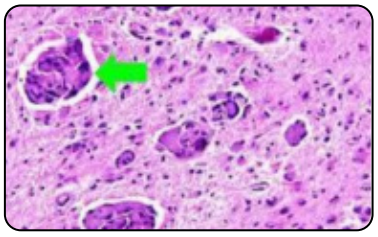
  Enzyme → Sphingomyelinase



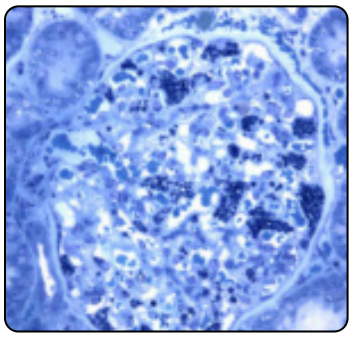
- Gaucher's Disease

  Enzyme → ß- Glucocerebrosidase

→ Globoid cell



- Krabbe's Disease
  → Defect in GALAC gene lead to demyelination
  → Enzyme - B- Galactocerebrosidase
- Fabry's Disease
  → Enzyme - x- Galactocerebrosidase
  Intracellular accumulation of inclusion bodies with a myelin like app



- Tay Sach's Disease
  → Nerve in the brain are destroyed
  → Enzyme - Hexosaminidase-A
- Sand Hoff's Disease
  → Enzyme - Hexosaminidase-A&B



D. Nucleotid's

- Lesh Nyman Syndrome
  C/F - Nail & Lip biting
  → Enzyme - HGPRT

- Amino Acid
  1. Tyrosine
  2. Tryptophan
  3. Histidine
  4. Arginine
- Inhibitor
  1. Fluoride
  2. Barbiturate and Rotenone
  3. Malonate
  4. Co, CN, H2S
  5. Oligomycin
- Special Product

  T3, T4, Catecholamine, Melanin

  Vit B3, Serotonin, Melatonin

  Histamine

  Nitric Oxide

  Enolase

  Complex-1

  Complex-2

  Complex-4

  Complex-5
- Classification of amino acids
  1. Aliphatic Amino Acid - G-Vali

     Glycine, Valine, Alanine, Leucine, Isoleucine
  2. Hydroxyl Group Containing Amino Acid - S.T

     Serine, Threonine
  3. Sulphur Containing Amino Acid - M.C

     Methionine, Cysteine
  4. Acidic Amino Acid - AA- GG

     Aspartic Acid, Asparagine, Glutamic Acid, Glutamine
  5. Basic Amino Acid - HAL

     Histidine, Arginine, Lysine
  6. Aromatic Amino Acid - PTT

     Phenylalanine, Tyrosine, Tryptophan
  7. Essential Amino Acid -T.V till 8pm

     Tryptophan, Valine, Threonine, Isoleucine, Leucine, Lysine, Phenylalanine, Methionine

8. Semi Essential Amino Acid - HA
   Histidine, Arginine
9. Ketogenic Amino Acid - LL
   Leucine, Lysine
10. Glucogenic and Ketogenic Amino Acid - PITT
    Phenylalanine, Isoleucine, Tryptophan, Tyrosine
11. 21" Amino Acid
    Selenocysteine
12. 22" Amino Acid
    Pyrrolysine

- Miscellaneous

1. Uncouplers - 2, 4 - DNP, Aspirin
2. Lipoproteins with highest Cholesterol - LDL
3. Lipoprotein with highest Triglyceride - Chylomicron
4. Nature of enzyme - Apoenzyme + Coenzyme
   = Holoenzyme
5. Competitive Inhibition - Vmax - Same
   - Km - ↑
6. Non-Competitive Inhibition - Vmax - ↓
   - Km - Same
7. Protein folding & unfolding is done by
   – Chaperon

- Vitamins

→Fat Soluble – A, D, E, K

→Water Soluble - B & C

1. Vitamin A

Active Form - Retinol

Deficiency - Bitot spot

2. Vitamin D

Active form of D3 - 1, 25 – dihydroxy-D_3

Deficiency

Children → Rickets

Adults → Osteomalacia

3. Vitamin E

Active Form - Tochopherol

Deficiency - Inflammation

Cancer

Cellular Damage

4. Vitamin K

K1 - Phylloquinone

K2 - Menaquinone

K3 - Menadione

Deficiency - Bleeding disorder

5. Vit B1 - Thiamine

Deficiency - Wernicke's – Korskoff's Syndrome

- Beri Beri

6. Vit B2 - Riboflavin

Deficiency - Glossitis

Neural Dysfunction

Dermatitis

7. Vit B3 - Niacin

Deficiency - Pellagra

> Dementia

> Diarrhoea

> Dermatitis

8. Vit B5 - Pantothenic Acid

Deficiency - Burning Feet Disease

9. Vit B6 - Pyridoxine

Deficiency- Anaemia

10. Vit B7 - Biotin

Deficiency – Hairloss

11. Vitamin B9 - Folic Acid

Deficiency - Megaloblastic anemia

- Neural Tube Defect

12. Vit B12 - Cobalamin

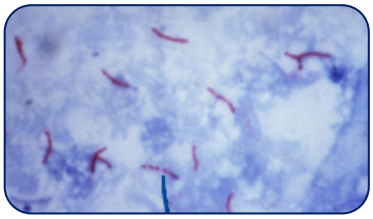
Deficiency - Megaloblastic anemia

- Sub acute combined degeneration of spinal cord

MICROBIOLOGY

- **Bacterial Smear Staining**
  - →Iodine = Mordant Fixe
  - →Acetone = Decolouriser
  - →Safranin = Counter stain
  - Thick Peptidoglycan = GP = Purple to Violet
  - Thin Peptidoglycan = GN = Pink
- **Acid Fast Staining =**

  Use = M. Tb = Z.N Stain

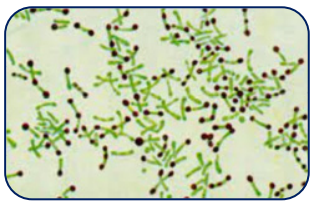


Blue Background

Pinkish-Red

- **Albert Stain**

  Use = C. Diphtheria (Green colour)



→Chinese Letter Pattern

- **Bipolar Stain**



→ Safetypin Appearance

Use = 1. Yersenia Pestis = Disease → Plague

Doc = Streptomycin

2. Burkholderia Malai = Leads to Glander's DS

3. Hemophilus Ducreyi = Leads to chancroid (Painful)

- Capsule =

Capsulated Organism = P - Pneumococcus, Pseudomonas
K - Klebsiella
B - Bordetella Pertusis → Leads to whooping/ 100-day cough
Bacillus Anthracis
C - Clostridium Perferinges/ Welchi
↳ Leads to gas gangerene
- Cryptococcus Neoformans → M.C.C. of Meningitis in HIV +ve
Most of the capsules are made up of Polysacchride except Bacillus Anthracis made up of Polypeptide

◆ Rxn =

1. Quellung Reaction = Pneumococcus
2. Mc Fadyean's Reaction = B. Anthracis

◆ Function of capsule =

1. Avoid opsonisation & prevent phagocytosis
2. Biofilm - Helps in attachment
- Helps in antibiotic resistance

◆ Test for detection of Polypeptide capsule

Latex Agglutination Test
Negative Staining
Mc Fadyen's staining

◆ Fimbria

Function- attachment
present in all bacteria causing UTI
Test to detect: Heme agglutination

◆ Flagella

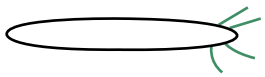
Thick Bacteria (M.C.) - Seen under Field/Light microscope
Thin Bacteria- phase contrast microscope/dark field
Example Borelia
Leptospira
Treponema

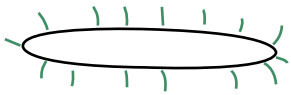
3. Motility
= Through flagella

Arrangement Of Flagella

1. =  = Monotrichous = Vi - Vibrio Cholerae
P - Pseudomonas

2. =  = Amphitrichous = A - Alcaligens
C - Campylobacter Jejuni

3. =  = Lophotrichous = Spirillum Minus

4. =  = Peritrichous = Proteous
Salmonella Typhi
E. Coli

Motility

1. Tumbling = Listeria Monocytogens
2. Darting = V. Cholerae, Campylobacter Jejuni
3. Swarming = Proteus
4. Gliding = Mycoplasma
5. Twitching = Trichomonas Vaginalis
6. Falling Leaf like = Giardia Lambia
7. Rotatory = Balantidium Coli
8. Cork screw = Spirochetes

◆ Culture Media

1. Simple Media =
Nutrient Agar
Consist of peptone & meat extract
2. Complex Media =

(Solid)	(Liquid)
A. Enriched Media • Blood Agar • Chocolate Agar • Loffler's Serum Slope	B. Enrichment Media • Selenite F Broth • Alkaline Peptone Water

3. Selective Media
   - TCBS = Thiosulpnate citrate bile salt sucrose
   - MSA = Mannitol salt agar
   - Potassium Tellurite Agar

4. Transport Media
   A. Venkat Rama Krishan Media → V. Cholare
   B. Cary Blair → V. Cholare
   C. Pikes Media → S. Pyogens
   D. Stearts & Amie's Media → Meningo & Gonococci

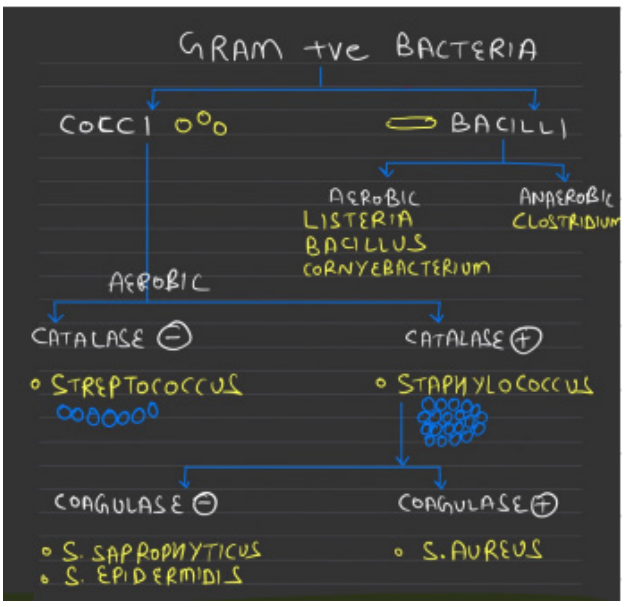
Anaerobic Media =

A. RCM = Robertson cooked meat media

Use = Clostridium

If meat pieces → Pink
= Clostridium Per Fringes

If meat pieces → Black
= Clostridium Tetani



A. GRAM +ve COCCI

CONS = Cogulase Negative Staphylococcus

1. S. Epidermidis

   +nt on normal skin flora

2. S. Saprophyticus

   Can cause UTI in young female

   (But overall Mcc is E. Coli)

COPS = Cogulase Positive Staphylococcus

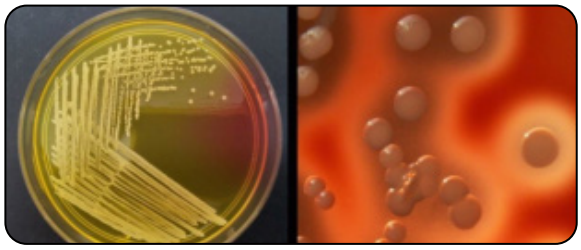
1. Staph Aureus

   Catlase +ve & Coagulase +ve

   Purple → Violet grape like cluster +nt

Cx Media =

- Nutrient Agar → Golden Yellow pigmentation
- Blood Agar → β- Hemolytic colonies



→ Best selective media = Mannitol Salt Agar

→ Virulence factor of S.Aureus

1. Thick layer of Peptidoglycan
2. Teichoic Acid (GP)
3. Clumping Factor

→ S.Aureus can cause food poisoning D/t ontaminated milk & meat etc .

Disease caused by Staph Aureus

→ Cellulitis

→ Carbuncle

→ Native valve endocarditis

→ I.V drug abusers = Right sided valve endocarditis

→ Sub-Dural/ Epidural Abscess

MRSA = Methicillin Resistant Staphylococcus Aureus

→DOC = Vancomycin

S.E = Red Man Syndrome

→Best method to avoid nosocomial spread of MRSA = Hand Washing

Streptococcus = (G. Positive COCCI)

→Catalase -ve COCCI

α	β	γ (Gamma)
• S.Pneumoniae • S.Viridans	• Group.A- Strep. Pyogens • Group-B-Strep. Agalactiae	• Enterococcus Faecalis • E. Faecium

S. Pyogens =

→Violet colour COCCI arranged in chains

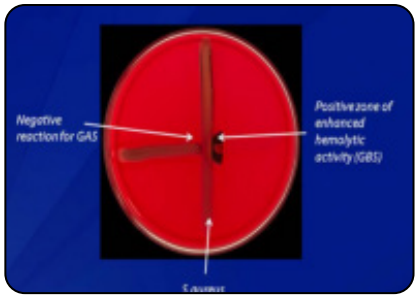
- Cx Media = CVBA [Crystal Violet Blood Agar]
- Transport Media = Pikes Media
- Disease = 1. Acute Pharyngitis
  2. Tonsillitis
  3. Cellulitis
  4. Impetigo [Honey coloured crust]
  5. Necrotizing Fascitis
  6. Scarlet Fever

↳ Strawberry Tongue

GBS = Group B Streptococcus also has S.Aglactiae

Mcc of neonatal meningitis in world

→ Camp Test +Ve



Enterococcus

2 Types

A. Faecalis B. Faecium

+nt in normal flora of gut

M.C.- Surgical wound contamination

DOC = Penicillin + Aminoglycoside or Ampicillin

2nd Line = Vancomycin

VRE [Vancomycin Resistant Enterococcus]

↳ DOC = Linezolid/ Daptomycin

Streptococcus Viridance

> S. Mutans → Dental Caries

> S. Viridance→ • Sub-Acute bacterial endocarditis

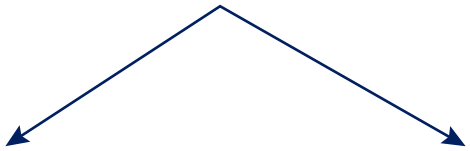
• Late onset prosthetic valve endocarditis > 1 yrs

Duke's criteria is used for Endocarditis

Streptococcus Pneumoniae =

→ Catlase -ve
→ Capsulated = Quellung Reaction
→ Bio +ve
B → Bile Solubility
I → Inulin Fermentation
O → Optochin Sensitivity
→ Disease =
- Pneumonia = MCC of [Community acquired pneumonia]
- Conjunctivitis
- Sinusitis
- Meningitis

CAP = Community Acquired Pneumonia



Typical (+nt)
- MCC = Pneumococcus Pneumoniae Streptococcus

Atypical (-nt)
- MCC = Mycoplasma Pneumoniae

- MCC of meningitis in all age groups is

Pneumococcus except neonates

↳ S.Aglactiae (GBS)

DOC = Penicillin

- Important causes of typical pneumonia

↳ Typical pneumonia consolidation +nt

↳ Pneumococcus pneumonia

↳ MCC of Pneumonia

↳ Staphyllococcus aureus

↳ MCC of Nosocomial Pneumonia

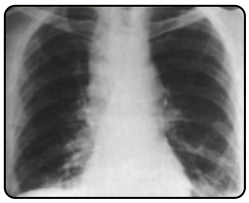
↳ Pseudomonas
↳ MCC of pneumonia in Cystic fibrosis patient

B. GRAM Positive Bacillus [GPB]

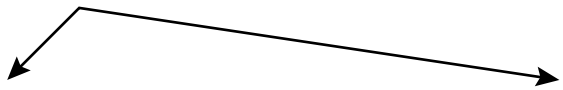
1. C. Diphtheria
→ on Albert Stain = Chinese Letter Pattern
→ Biotypes of C. Diphtheria
a. Gravis → Daisy Head appearance
b. Intermedius → Frog Egg Appearance
c. Mitis → Poached Egg Appearance
→ Disease = Diphtheria
C/F • Pseudomembrane = Greyish white (bleeds on removal)
• Bull's neck +nt due to Lymph Adenopathy
→ Complication = • Myocarditis (Max Death)
→ Confirm Dx by = Potassium Tellulite Agar
→ Rx →• TOC – Anti-diptheric serum
• Initial Stage = Penicillin/ Erythromycin

2. Bacillus Anthracis =
→ Capsulated Polypeptide
→ MC Fadyean Reaction
→ Appearance = M - Medusa Head appearance
I - Inverted Firtree appearance
S - String Of Pearl appearance
Bamboo - Bamboo Stick appearance
→ Disease

a. Cutaneous Anthrax ↳ Lead to hide porter's disease	b. Pulmonary Anthrax ↳ Wool Sorter Ds.

c. B. Cerus =



Emetic
→ Chinese Fried Rice

Diarrheal
→ Contaminated Meat & Milk Product

Cx Media = B.Cerus = MYPA

3. Clostridium

→Spore forming

Cx Media = RCM (Robertson Cooked Meat Media)

↳ Cl. Perferenges =

→Gas gangrene

→Thick brick shaped GPB

↳ Cl. Tetani = —○ (Drum stick appearance)

→Tetanus (8th day disease)

↳ I.P = 6-10 days

↳ S/S = • Lock Jaw

Opisthotonos Posture

↳ Cl. Botulinum =

A → Enterotoxin

↳Canned food poisoning

S/S Dysphagia, Diplopia except constipation

B → Neurotoxin

↳ Botox treatment for wrinkles

↳ Eye surgeries

Cl. Difficle =

↳ Lead to CDAD = Cl. Difficle associated diarrh

DOC = Oral Vancomycin

If not treated on time it will lead to Pseudo Membranous Colitis

4. Mycobacterium Tuberculosis =

→ Acid fast

→ In children's lower lobes are involved

- Ghon's Focus + Hilar Lymphadenopathy → Ghon's complex
- Calcified Hilar Lymphadenopathy → Rankes complex

→ Post Primary TB, Upper lobes are involved

- Subpleural focus → Simmon Focus
- B/L Infra Clavicular Lesion → Assman's Focus

→ Skin TB

- Mc Primary Lesion → Lupus Vulgaris
- 2nd Mc Lesion → Scrofuloderma

→ GI TB

- Mc Site → Ileocecal Junction

→ Bone Tb → Pott's Spine

Dx =

1. Sputum Examination
   a. ZN staining + X-Ray Chest
   b. LJ media
2. MTB PCR Method
   [GENE xpert (Detection of rifampicin resistance)]
3. CBNAAT
   [MTB PCR + Multi drug resistance detection]
4. QuantiFERON TB gold assay
5. Tuberculin test/ Mantoux test

Rx = BCG –Vaccine

C. Gram Negative COCCI

Meningococci =

→ Capsulated

→ Lens shaped

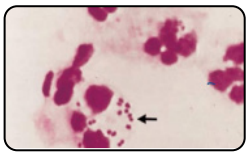
→ Most infective serotype - A, B, C

→ Spread → Droplets infection

Disease = Meningitis

S/S → Fever, Vomiting, Headache, Neck rigidity, Rash

Dx = 1. CSF Examination
2. Culture



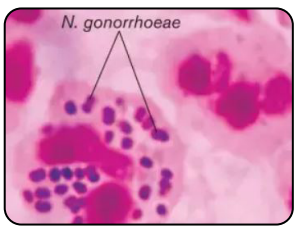
Gram staining shows -pink coloured COCCI

→ Most dreadful complication = Water house Friderichsen syndrome

Rx = Doc for carriers = Ciprofloxacin
Doc for cases = 3rd generation Cephalosporins

- Gonococci

Disease = Gonorrhea =
→ Male → Urethritis
→ Females → Cervicitis
→ Newborn → Ophthalmia Neonatrum



- Complication =
  1. Water Can Perineum
  2. Fitz Hugh Curtis Syndrome
  3. Poly Arthritis

Gram Negative Bacillus

1. E. Coli [catalase +ve & oxidase - ve]

I - Inodle	Positive
M - Methyl Red	Positive
Vi - Voges Proskauer	Negative
C - Citrate	Negative

Disease Caused =

A. U.T.I

B. Diarrhoea

EPEC → Cause paediatric diarrhoea

EIEC → Resemble shigellosis & sereny test +ve

ETEC → Cause traveller's diarrhoea

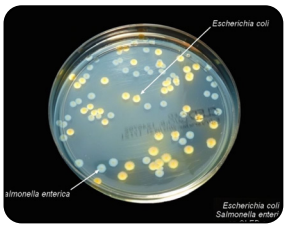
EHEC → Cause hemolytic uremic syndrome in children

EAEC → Cause persistant type of diarrhoea

C. Neonatal Meningitis

MCC of Neonatal Meningitis





2. Klebsiella = [Urease +ve & capsulated]

I - Indole	= Negative
M - Methyl Red	= Negative
Vi - Voges Proskauer	= Positive
C - Citrate	= Positive
Organism	Disease

1. Klebsiella Pneumonia → Typical Pneumonia
2. Klebsiella Ozaena → Atrophic Rhinitis
3. Klebsiella Rhinoscleromatis → Rhinoscleroma [Woody Nose]

3. Salmonella Typhi
   Disease = Typhoid Fever
   Transmission = Faeco Oral route
   C/F = Step Ladder Pattern fever
   Rose Spots seen
   Longitudinal Ulcers seen
   Pea Soup Diarrhoea seen
   Culture MEDIA = Wilson & Blair media (best)
   Lab Diagnosis = B - Blood culture – 1st week
   A - Agglutination test – 2nd week
   S - Stool culture - 3rd week
   U - Urine culture – 4th week
   Doc = 3rd generation cephalosporin

4. Shigella (Catalase -ve & non motile)
   Secrete Shiga toxin by S. Dysentriae
   Mcc of bacillary dxsentry in India → S. Flexneri
   Enrichment Media - Selenite F-Broth

5. Vibrio Cholerae = [Catalase +ve, oxidase +ve, indolase +ve]
   Lead to - Rice Watery diarrhoea
   Non-Inflammatory diarrhoea
   Cholera Toxin → Mimic arsenic metal poisoning
   Culture Media →
   Transport Media - Venkatraman Ramakrishan
   Enrichment Media = Alkaline peptone water

Selective Media = 1. TCBS (Thiosulphate citrate bile salt sucrose)

6. Haemophilus Ducreyi
   → Cause chancroid [Painful genital ulcer]
   → issue Biopsy + Staining shows Tram track appearance or school of fish appearance

7. Yersinia Pestis
   → Disease caused → Plague [I.P = 2-7-Days]
   → Culture media → Ghee Broth
   → Doc = Streptomycin

8. Legionella Pneumophila
   → Spread = Aerosol route from centralised AC plant
   → Disease = • Pontiac fever
   • Leggionare's pneumonia
   → Culture Media = BCYE – [Buffered Charcoal Yeast Extract]

9. Treponema Pallidum

Disease = Syphilis [9-90 Days]

Stages of Syphilis

1° Stage = Hard painless chancres seen

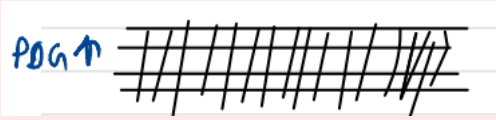
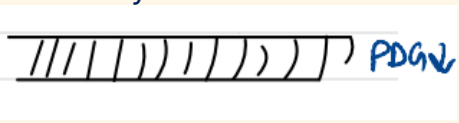
2° Stage = Rash over palm & soles
Condyloma lata lesion are seen

3° Stage = Gumma Satin Lesion

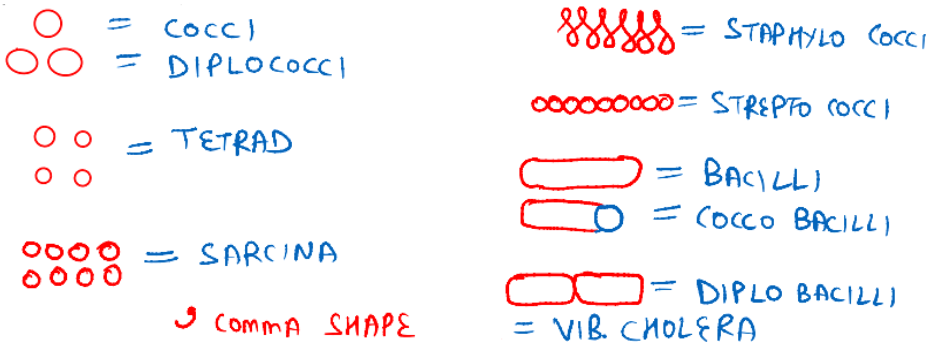
DX = MC = TPHA - Treponema Pallidum Haemagglutination assay

10. Mycoplasma =
   → Smallest bacteria
   → Cell wall -nt & Cell membrane +nt
   ↳ Culture media = PPLO Agar [Fried egg colonies]

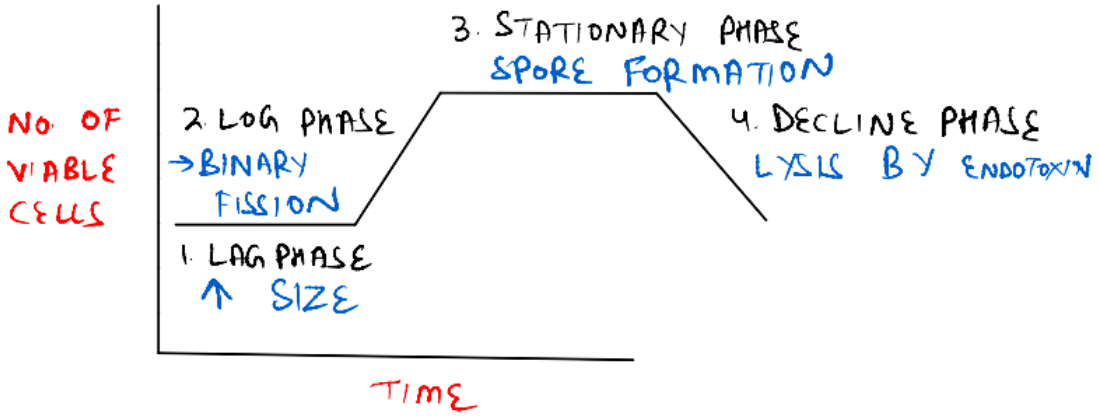
Bacteria	Virus
• Large • RNA + DNA	• Small • RNA OR DNA

• Multiply by = Binary fission • Prokaryote	• Multiply by = Infective cycle • Neither Prokaryote nor Eukaryotes
Gram +ve → 100 lyers  → Teichoic acid +nt (Act like fenicol) → Outer membrane -nt	Gram - Ve → 1-2 Layers  → Teichoic acid -nt +nt

Shape



Bacterial Growth Curve =



Sporocidal Agents =

→ Ethylene Oxide = Heart Lung Machine

→ Formal Dehide = OT, Ward

→ Glutaraldehyde= Bronchoscope, Endoscope, Laryngoscope

→ Isopropyl Alcohol = Thermometer

→ Phenol = Dettol → Chlorxylenol

Savlon → Chlorhexidine + Citrimide

Blood Spill → Na Hypochloride

- **Sterilization =**

A. Dry Heat Sterilization

[Do-oxidative damage]

1. Incineration = Use = Hospital waste Temperature = 1200°C	2. Hot air oven Use = Glassware, Cotton swab, etc Temperature = 160°C Time = 2hrs

B. Moist Heat Sterilization

[Coagulation & denaturation of protein]

< 100°C	100°C	>100°C
1. Pasteurization 2. Inspissation	3. Boiling 4. Tyndalisation	5. Autoclave

1.Pasteurization of milk

a. Flash method	b. Holder's method
→ 72°C → 15-20 sec	→ 63°C → 30 Min

2. Inspissation

→ 80 - 85°C

→ 20-30 minutes for 3 days

3. Tyndallization

→ 100°C × 20-30 min 3 days

4. Autoclave

→ 121°C 15min 3 days

Used for – Culture Media except LSS & LJ media

↓

Method used: Inspissation
Tyndalisation

Mycology

- **Classification based on location**

1. Sub-Cutaneous Mycosis =



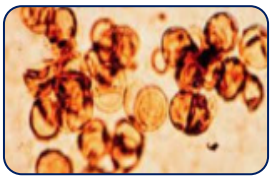
A. Mycetoma/ Madura foot

→ Triad = SC nodule

Multiple discharging sinus

Sulfur granules

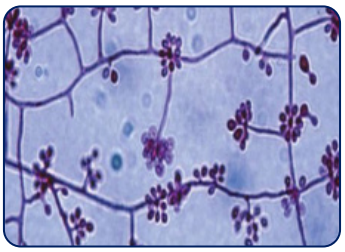
Actinomycetoma	Eumycetoma
→Bacterial →White Granules on x-ray = Osteolytic vvvLesion	→Fungal →Black Granules →Osteosclerotic Lesion



B. Chromoblasto Mycosis

Sclerotic bodies are seen in tissue body (brown colour) → Copper penny appearance

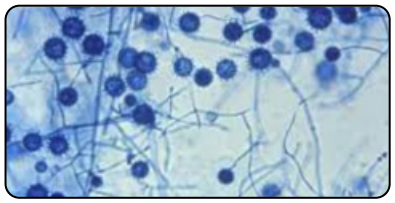
C. Sporotrichosis/ Rose Gardner's disease



→ Dimorphic Fungus
→ Caused by Sporothrix Schenki
→ Spread by Thorn Prick
→ Flower shape appearance
→ Cigar shape appearance



D. Rhinosporidiosis - caused by Rhinosporidium Seeberi



2. Deep Mycosis

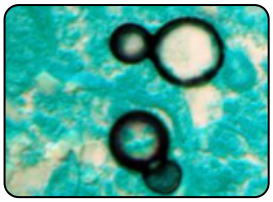
A. Histoplasmosis [Darling disease]

→ Caused by Histoplasma Capsulatum

→ Aerosol transmission of Pigeon Faeces occur

→ Mimic TB

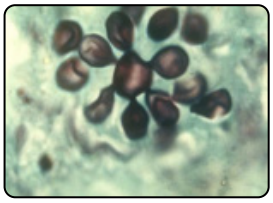
B. Blast Mycosis/North American disease



→ Caused by blastomyces dermatitis

→ Look like figure of 8 appearance

C. Paracoccidiomycosis/ South American disease



→ Caused by Paracoccidiodes Brasilensis

→ Mickey mouse or pilot wheel appearance

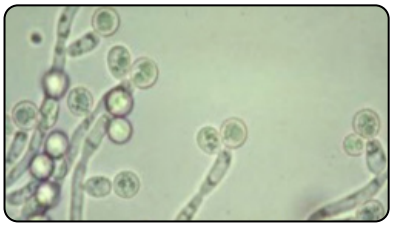
D. Candida Albicans

Shows → Reynold Braude phenomenon

Disease =

→ Oral thrush

→ Vaginal candidiasis



Lab diagnosis

→ On SDA = Creamy, pastry colony

E. Cryptococcus Neoformans



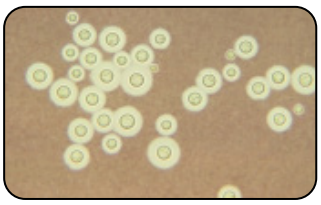
→ Aerosol transmission through Pigeon Faeces

→ Mc-cause of meningitis in HIV +ve

Pt. is → Cryptococcus Neoformans

Dx = → Indian ink staining

→ On SDA



DOC = Amphotericin B. + Flucytosine

3. Superficial Mycosis

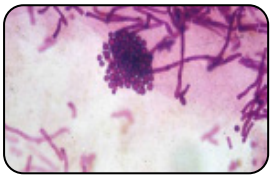
A. Tinea Versicolor = Scales +nt on trunk

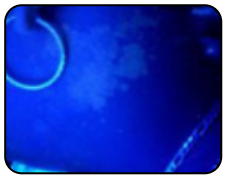


Lab Diagnosis =

→ caused by Malassezia Furfur

- Scales + 10% KOH → Spaghetti & meat ball appearance
- Scales + Wood Lamp → Golden Yellow Fluorescence





Rx = Selenium Sulphide

B. Piedra =

→ Black Piedra → By Piedra Hortae

→ White Piedra → By Trichosporon Beigel



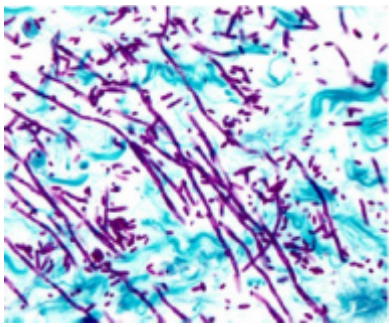
4. Occupational Disease

A. Farmer's Lung =

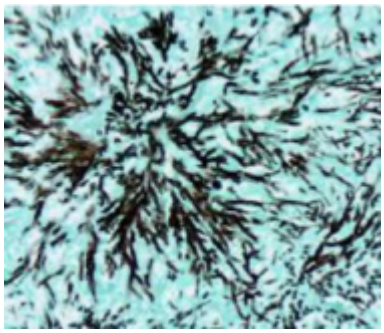
→ D/T Hay dust → Thermo Actinomyces Vulgaris
Micromonospora Faenii

→ Type III - Hypersensitivity

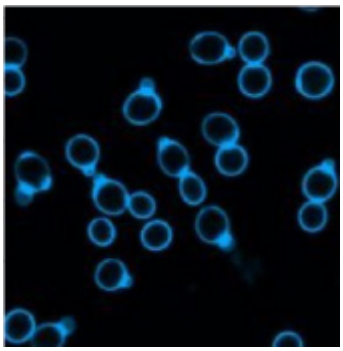
- **Stain in Fungal Infection:**



PAS



Gomori Methamine Silver Stain



Calco Fluor Stain

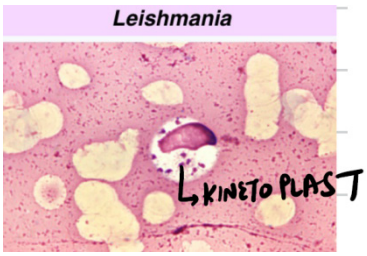
- **PROTOZOA**

  Leishmania

  → Transmitted by Sand Fly

  → Disease: Leish Maniasis

  → Dx: Rk39 Ag +ve



  Trypanosomia Cruzi

  → Transmitted By Reduvig Bus

  → Dissase: Chagas Disease

  Trypanosomia Bruci

  → Transmitted By Tse - Tse Fly

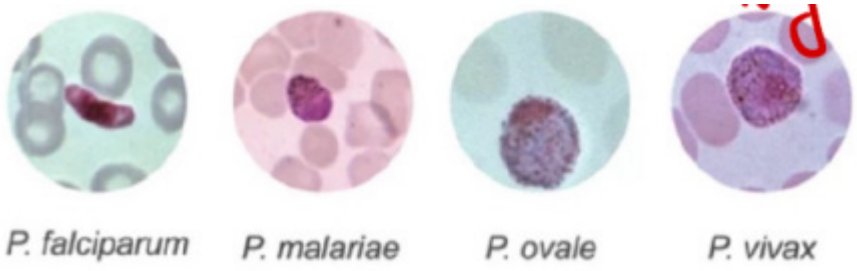
  → Disease: Sleeping Sickness

- **Coccidian Parasite**

  C/P: HIV +ve + Diarrhoea

Feature	Cryptosporidium	Cyclospora	Isospora
Size Shape Acid Fast Rx	4-6µ Round + Nitazoxanide	8-12µ Round + Cotrimoxazole	24-36µ Oval + Cotrimoxazole

P. Vivax	P. Falciparum
↓ Malaria Ag- PLDH	↓ Malaria Ag- PLDH HRP2



- Toxoplasma

  ↓

  → Toxoplasmosis

  → Spread through feco-oral route

Virology

- All DNA viruses are double stranded except PARVO virus
- All RNA virus are single stranded except REO virus

- DNA Virus =

H2 → Herpes
→ Hepadena
A → Adeno
P → Parvo
P → Papova
P → Pox
y

Shape of Virus
→ Bullet shaped → Rabies Virus
→ Rod shaped →Tobacco Mosaic Virus
→ Brick shaped → Pox Virus
→ Space vehicle → Adenovirus
→ Filamentous → Ebola Virus

- Inclusion bodies =

A. Intra Cytoplasmic Inclusion Bodies
→ Negri Bodies → Rabies
→ Guarneri Bodies → Vaccinia
→ Paschen Bodies → Variola
→ Bollinger Bodies → Fowl Pox
→ Henderson Peterson Bodies → Molluscum Contagium

B. Intra Nuclear Inclusion Bodies

1. Cowdry Type A → Herpes virus → Yellow fever virus [Torres bodies also seen}	2. Cowdry Type B → Adenovirus → Polio virus

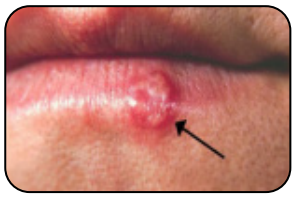
DNA Virus =

1. Herpes =

→ Family =

α-Virinae	β-Virinae	γ-Virinae
• HSV-1 • HSV-2 • HHV-3 → Varicella Zoster [Chickenpox Virus]	• HHV-5 – Cytomegalo Virus • HHV-6 • HHV-7	• HHV-4 • HHV -8- [Kaposi Sarcoma Causing Virus]

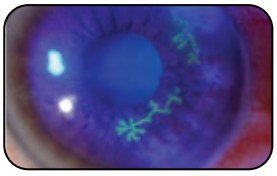
HSV = → HSV – 1 [Above waist]
→ HSV - 2 [Below waist]



A. Mucosal involvement

→ Buccal Mucosa Involved → Acute Gingivo Stomatitis
→ Mc Recurrent Herpes Lesion → Herpes Labialis

B. Ophtha involvement
→ Acute Dendritic ulcers
→ Rx - Topical Acyclovir

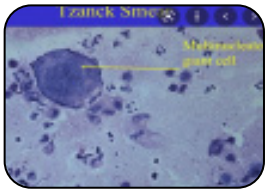


C- Genital involvement
→ B/L Vesicular, Erythematous Lesions

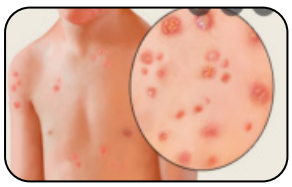


Note = How to prevent HSV2 inf. to baby ?
Cesarean section done

Diagnosis = PCR
Elisa
Tzanck Smear - Tzank cells



HHV-3 = Varicella Zoster/ Chicken Pox Virus
→ Port of Entry → Respiratory Tract/ Conjunctiva
→ Centripetal rash +nt, dew drop appearance & pleomorphism seen



→ I.P = 14-16 days
→ Mc-Complication = 2° bacterial inf



Reactivation of Herpes Zoster = Shingles
→ Dermatomal involvement

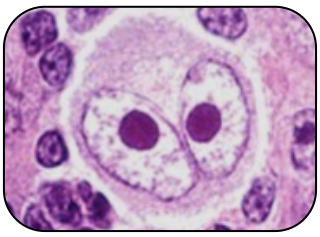
Types =
A. Herpez Zoster [Ramsay Hunt Syndrome]
→ Bell's palsy
→ Geniculate Ganglion involved

B. Herpez Zoster Ophthalmicus

→ Ophthalmic branch of trigeminal nerve involved

- HHV-5 – Cytomegalo Virus

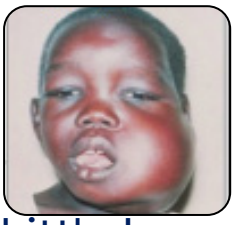


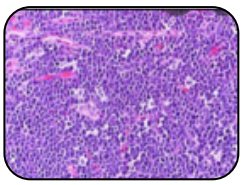


→ Owl eye appearance seen

Most common C/F = Hepato Splenomegaly

tumor of jaw





starry sky appearance

- EBV =

Disease – 1. Burkitt's Lymphoma

2. Naso Pharyngal Carcinoma

3. Oral Hairy Leukoplakia

4. Infectious Mononucleosis/ Kissing Disease

- HHV-8 = Kaposi Sarcoma

→ Common in HIV +ve pt

→ Vascular Tumor



- HHV 6 & 7 →

→ Causes Exanthem Subitem/ Roseola Infantum

2. Pox Virus =

- Molluscum Contagiosum

→ Pinkey Perly dome shaped papule

→ On Biopsy Henderson Peterson body seen

→ Rx = Podophylin + Cryo surgery



3. Parvo Virus =

Most imp. = B-19

Associated with = Erythema Infectosom/ 5th day D



Slapped cheek appearance seen

4. Adeno Virus [Space vehicle]
   → Non-Enveloped Virus
   → Types 3, 4, 7, 14 → Lead to swimming pool conjunctivitis
   → Types 8, 19, 37 → Epidemic Kerato conjunctivitis

5. Human Papilloma Virus =

→ HPV 1 & 2	→ HPV 6
• Cause common warts 	• Causes genital warts [Condyloma acuminate] 

HPV 16 = Cancer Cervix



RNA Virus:

1. Polio Virus =
   - Route = Faeco oral contamination
   - Paralysis - Hand knee gait seen
     [Anterior horn cells are damaged]
   - Dx = Elisa/ PCR

2. Coxsackie Virus =

Coxsackie Virus A	Coxsackie Virus B
• Hand Foot Mouth Dis. • Aseptic Meningitis • Flaccid paralysis in mice	• Hand Foot Mouth Dis. • Aseptic Meningitis • Spastic paralysis in mice

3. Myxoviruses

A. Orthomyxoviridae

→ DIS = Influenza A, B, C

→ H5N1 → Bird Flue

→ H1N1 → Swine Flue

B. Paramyxoviridae

a. Para Influenza =

→ Type 1 & 2 → Croup Disease

→ Type 3 → Lower respiratory tract infection

b. Mumps = I.P = 2-3 wks.



- Mc Presentation = B/L Parotitis
- Complications

U/L Orchitis

Oophoritis

c. Measles



- Most characteristic feature → Koplik spot
- Late complication = SSPE (Sub acute chronic pan encephalitis)
- IP = 10-14 days
- Infectivity period - 4 days before & 5 days after rash
- Vaccine - MMR

d. Rubella/ German Measles =

- Infectivity Period = 1 week before & 1 week after rash
- IP = 2- 3 Weeks



→ Forcheemier spots on uvula

→ Blue berry mufin rash

- Congenital Rubella Syndrome

C - Cataract
H – heart disease
D - Deafness (SNHL)

4. Rabies Virus =

- SS RNA virus
- Pt. develop hydrophobia
- Negri bodies +nt
- Post Exposure Prophylaxis

→ 0,3, 7, 14, 28

- Pre Exposure Prophylaxis

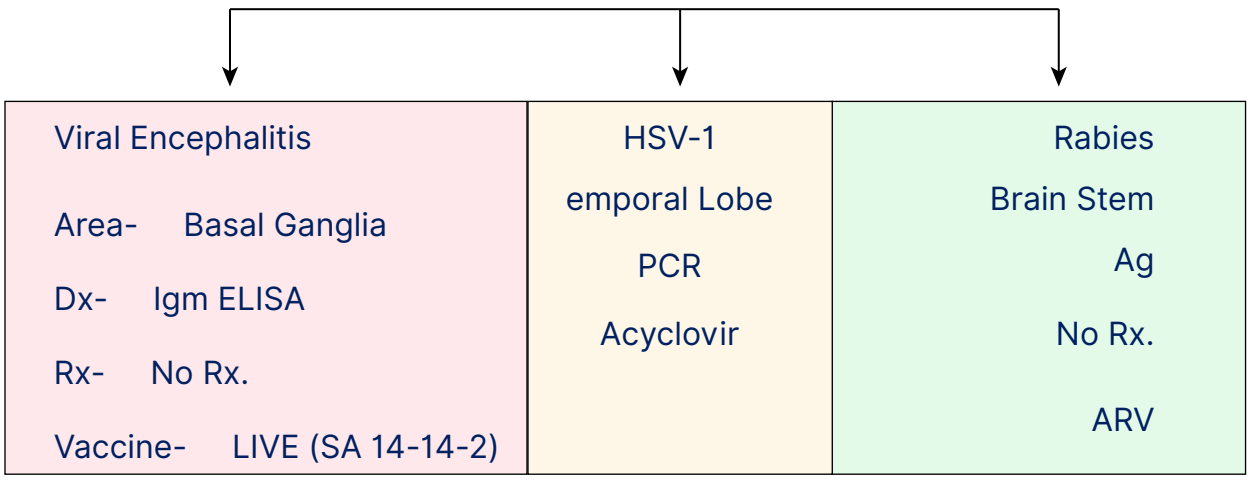
→ 0, 7, 21 or 28

Diagnosis

- Ante Mortem = Corneal smear, nape of neck H.F
- Post Mortem = Negri body seen in hippocampus

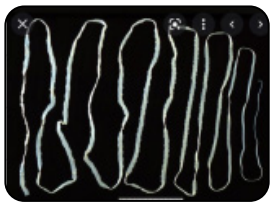
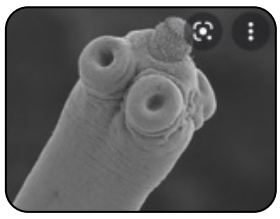
Dose of rabies immunoglobulin - Human Rabies 20 uni/Kg.

- **Viral Encephalitis**



Helminthology

1.Cestodes =

- Tape like
- Sucker with hook
- Body cavity absent

Types =

A. Taenia Solium



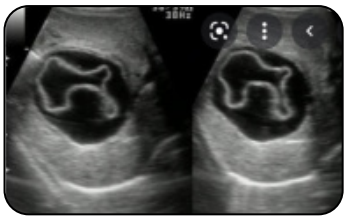
- Definitive host = Man
- Intermediate host = Pig
- Tape like
- Cause due to uncooked pork

S/S Taeniasis
(Diarrheal Episodes -+nt)

B. Taenia Saginata

- Definitive host = man
- Intermediate host = Cattle
- Tape like
- Cause due to uncooked beef

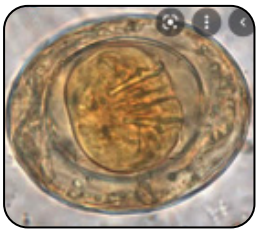
C. Echinococcus granulosus/ dog tapeworm



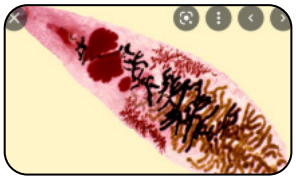
- Forms hydatid cyst
- Definitive host = dog
- Intermediate host = sheep/ man

D. H.Nana/ Dwarf tape worm

- Eggs are ingested with contaminated food & water
- Definitive host = man
- Intermediate host = man



2.Trematodes/ Flukes



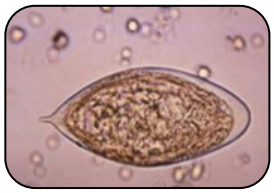
- Leaf like
- Sucker's without hooks

3.Schistosomes

Single Intermediate Host → Snail

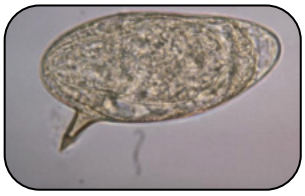
Types

A) S. Haematobium



→ Cause Sq.CC of urinary bladder

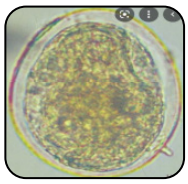
→ Cause hematurga



→ Egg has terminal spine

B) S. Mansoni

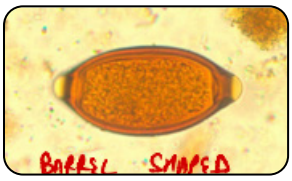
→ Cause swimmer itch & Katayama fever



→ Egg has side spine

C) S. Japonicum

→ Cause Katayama fever & Eosinophilic diarrhoea



→ Side hook present

4.Nematodes =

Large Intestine Nematode

A. Trichuris Trkhiura / Whip worm



- Coconut cake like rectum is seen
- Lead to iron deficiency anaemia

B. Entrobius Vermicularis/ Pin Worm



→ Presents with perianal pruritis

Small Intestine Nematodes

A.Ancylostoma Duodenal/ Hook Worm

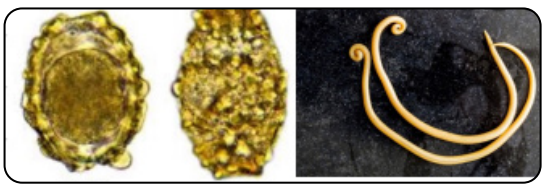


→ Penetration in the foot through skin



→ Anklostoma eggs

- Chandlers index used



B. Ascaris

- Eggs are ingested with contaminated food & water ipsum
- Can lead to Intussusception → Obstruction

Tissue Nematodes

A. Wucheria Bancrofi

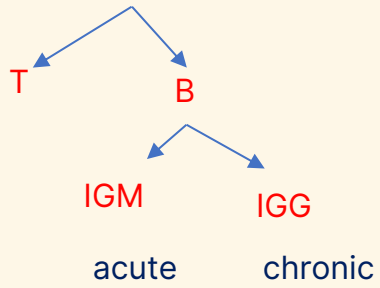


- Spread = Lymphatic
- Transmitted by = Culex
- Can lead to Elephentiasis & Hydrocele

Immunology

→1st Line defence = Skin, Mucous membrane

→ 2nd Line defence = Fever, Inflammation, Phagocyte

Innate Immunity	Acquired Immunity
→ +nt by birth	→ Comes after 1 infection
Components	
→ Dendritic cell → Natural killer cell → Acute phase protein → Normal resident flora	→ Classical Compliment pathway → Antigen presenting cell → Lymphocyte 
→ Takes minutes to develop → - Memory	→ Takes days to develop → - Memory

- Local Immunity = • G.I. Flora
  • Respiratory Flora } IgA
- Herd Immunity = Community
  Infected
  Vaccinated
- Antigen = Immunogen
- Any foreign substance which induces antibody formation
- Subtype = HAPTEN → • Incomplete antigen
  • Require protein for action

- Types Of Antigens =

A - Auto Antigen eg. • D - DNA • T - Thyro Globulin • C- Corneal tissue	B-Iso Antigen •Same species but diff. antigen eg. Humans - • Blood group • HLA type
C- Hetero antigen • Same antigen but diff. species eg. Mycoplasma = Human RBC Sheep RBC = EBV	D- Super Antigen • Multiple T&B cell stimulation eg. Staphylococcus

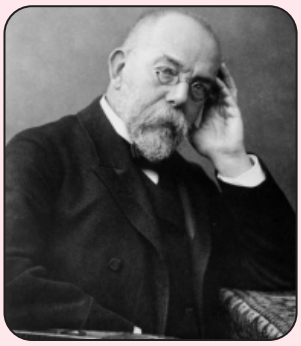
- **Antibodies =**
  - Ab are the immunoglobulin
    IgG = G1, G2, G3, G4
  - Only Ab which can pass through placenta except G2
  - Marker of chronic infection
  - IgG is opsonin – (Easy to digest)
    IgA = Dimer
  - Provide local immunity
  - +nt in all secretion except CSF
  - Mc Immunoglobulin def. is IGA
    IgM = Max. Molecular Weight
  - Pentamer molecule
  - 1st Ab to be synthesized in immune response
    IgD =
  - Marker of acute infection
  - +nt on β-cell
  - Helps in antigen recognition
    IgE =
  - Recognise Ab
  - Type 1 H.S rxn. mediator
  - Homocytotropic Antibody
    An Ab that has an affinity for mast cell

- Serological reaction Ag Ab interaction
  → Affinity = Intensity of attraction b/w the antigen & antibody
  → Avidity = Strength of bond after the formation of antigen & antibody complex
- Important Test =
  - Widal Test - For Salmonella Typhi
  - CAT - Cold agg. test - For mycoplasma
  - MAT - Microscopic agg. test - For leptospirosis
  - COOMB'S Test
    → Direct COOMB - For fetal Ab
    → Indirect COOMB - For maternal Ab
  - Schick Test - For Diptheria
  - Naglers Rx - For Cl.Perfringens
  - Wasserman Reaction - For Syphilis
  - Sabinfeldman Dye Test - For Toxoplasmosis
- Important Macrophage (Phagocytosis)
  - Blood - Monocyte
  - Bone - Osteoclast
  - Liver - Kuffer Cells
  - Lung - Alveolar Cells
  - Kidney - Mesangial Cells
  - Brain - Microglia
  - Skin - Langerhans
  - Spleen – Littoral Cells
  - Placenta – Haufber Cells
  - Tissue – Histocytes
- MHC = Major Histocompatibility Complex
  - +nt on 6p
  → MHC I = APC = CD8
  → MHC II = APC = CD4

- Interleukins =
  - → Il-1 - Inflammation, Fever
  - → IL-2 - NK Cell
  - → IL-4 - Class switching to IgG
  - → IL-5 - Class switching to IgA
  - → IL-8 - Chemoattraction
- Important Immunodeficiency Disorder
  1. Job Syndrome - ↑ IgE
  2. Chediak Higashi Syndrome - ↓↓ Melanocytes
  3. β Cell Deficiency - • × Linked Agamma Globunemia
     • Common Variable Immune-Deficiency
  4. T & β Cell Deficiency - Severe combined Immuno-deficiency
  5. T-Cell Deficiency - • DiGeorge Syndrome
     • Chronic Mucocutaneous Candidiasis

General Microbiology

Robert Koch
→ Father of modern M.B
→ Given Koch Postulates



Louis Pasture
→ Father of M.B
→ Vaccine Given
C - Cholera
A - Anthrax
R – Rabies





Antony Von Leeuwenhoek
→ Father of microscopy
→ 1st person who saw bacteria

- Gene Trans

1.Transformation = Naked DNA taken up from the environment by bacterial cells

2. Transduction = Use of bacteriophages to transfer DNA b/w cells.

3. Conjugation = Direct transfer of DNA from one bacterial cell. to another